BLIND-SPOT DECOMPOSITION AND THE GEOMETRY OF SYSTEM COLLAPSE*

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Abstract. We propose a geometric framework for detecting and controlling structural instability in coupled dynamical systems via energy-topology equivalence. A *Blind-Spot Decomposition Theory* (BSDT) energy functional [32] E_{BS} is shown to be topologically equivalent to the system potential Φ through a Morse-index equivalence: $\mathrm{ind}_{E_{\mathrm{BS}}}(X_c) = \mathrm{ind}_{\Phi}(X_c)$ at every critical point, with two-sided gradient bounds and homotopy equivalence of sublevel sets. This recasts instability detection as a control problem on a *critical manifold* $\mathcal{C}^*$ where adaptive friction $\gamma^*(X) = \alpha/\lambda_{\max}(D^2\Phi_{\mathrm{pair}})$ achieves marginal stability—formalising procyclicality in financial regulation. Two physics-based *System Mode engines* operationalise the theory with Lyapunov-certified convergence, fused topological scoring, and distribution-free conformal calibration. The framework is validated across three domains: FDIC banking data (6-quarter GFC lead, AUROC = 0.867, zero false alarms), ERCOT power-grid dispatch (+72 h lead, AUC = 0.9997), and three-dimensional incompressible Navier–Stokes via a 34-run GPU stress test (N up to 256^3, Re up to 62,832, NVIDIA RTX PRO 6000 Blackwell, 532 min wall time). All runs complete without blow-up: adaptive viscosity reduces peak vorticity by 1.77× at Re ≈ 1257; the BKM integral remains finite at Re ≈ 62,832 ($\|\boldsymbol{\omega}\|_\infty^{\max} = 132.2$, BKM = 465.4); twelve random-phase initial conditions confirm IC-independence; five distinct feedback laws produce identical dynamics (spread < 0.04%); and cross-correlations $\mathrm{Corr}(P, D) > 0.85$ demonstrate tight production-dissipation coupling across the full laminar-to-turbulent range.

Key words. systemic risk, Lyapunov stability, entropy production, Onsager gradient flow, phase transition, adaptive friction, Navier–Stokes regularity, enstrophy, BKM criterion, blind-spot decomposition, online convex optimisation, physics-based scoring, Lennard-Jones potential, fused topology, false-alarm-rate calibration

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1. Introduction.

The 2008 Global Financial Crisis demonstrated that systemic risk resides not in individual institutions but in the geometry of their interactions. Existing frameworks—CoVaR [2], SRISK [6], network contagion models [1, 8]—detect stress *after* it propagates. This paper proposes a geometric framework for detecting and characterising structural instability in coupled systems, and derives adaptive stabilisation as a direct consequence of the underlying energy-topology structure. The key insight is that a detection functional calibrated on normal-period data shares the same Morse-index landscape as the interaction energy—so detecting instability and controlling it become two views of the same geometric object.

Formally, an *equivalence theorem* links two seemingly unrelated objects: (i) a *detection* functional E_{BS} measuring deviation from historical normality, and (ii) the *interaction energy* Φ governing agent dynamics. Theorem C establishes that E_{BS} and Φ share the same Morse-index structure at every critical point, that their sublevel sets are homotopy equivalent, and that gradient norms are equivalent up to computable constants. This identification transforms the instability detection problem into a control problem: damping the system using $\operatorname{grad} E_{\mathrm{BS}}$ stabilises it because $\operatorname{grad} E_{\mathrm{BS}}$ and $\operatorname{grad}\Phi$ are geometrically aligned. The Morse-index jump from 0 to ≥ 1 provides a topological instability signal that is structurally invariant under perturbation and decoupled from the Lyapunov descent certificate used to guarantee iteration convergence.

A *critical manifold* $\mathcal{C}^*$ separates two dynamical regimes. Below $\mathcal{C}^*$, any fixed coupling suffices for stability. Above $\mathcal{C}^*$, no constant friction stabilises all configurations

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(Theorem 4.3). The system requires *adaptive* friction $\gamma^*(X)$ that tracks the spectral radius of the pairwise Hessian in real time. This phase-transition structure provides a rigorous dynamical-systems characterisation of procyclicality: time-invariant regulatory rules (such as Basel III's fixed countercyclical capital buffer) fail *structurally*, not merely in practice.

Contributions.. This paper makes nine contributions:

1. **Blind-Spot Decomposition Theory (BSDT).** Four deviation operators (δ_C: camouflage, δ_G: feature gap, δ_A: activity anomaly, δ_T: temporal novelty) define E_{BS}, a composite energy functional that decomposes structural instability into geometrically interpretable channels, calibrated entirely on normal-period data.
2. **Equivalence Theorem (Theorem C) and Entropy Structure (Theorem 3.13).** Morse-index equivalence $\mathrm{ind}_{E_{\mathrm{BS}}} = \mathrm{ind}_\Phi$ at every critical point, topological energy equivalence via homotopy of sublevel sets, two-sided gradient bounds, and decoupled stabilisation establish E_{BS} as a Lyapunov function for the potential flow (Proposition 3.8); the system is a generalised Onsager gradient flow with non-negative entropy production $\sigma(X) = \gamma(E_{\mathrm{BS}})\|\mathrm{grad}\, E_{\mathrm{BS}}\|^2 \geq 0$. This equivalence provides the mathematical basis for both detecting instability (via the Morse-index jump) and controlling it (via gradient-aligned damping).
3. **Critical manifold and phase transition (Theorem 4.3).** A spectral threshold determines where constant friction fails and adaptive friction succeeds, providing a rigorous dynamical-systems characterisation of why constant-parameter regulation fails in complex networks.
4. **System Mode engines (Section 6).** A Lennard-Jones Molecular Engine and a Gaussian/Coulomb Gravity Mode Engine share an identical post-simulation operator stack (Phase, Topological, KernelRKHS, Rank) fused via a Morse/Betti/UDL scoring architecture. A Hybrid Engine selects the optimal convex blend by cross-validated AUROC maximisation.
5. **Lyapunov v2 convergence guarantee (Proposition 6.1).** Three-part certificate: Armijo–ISS descent, La Salle invariance, and coercivity ensure convergence for all engines with $O(nk)$ per-step complexity.
6. **Online algorithm with $O(\sqrt{T})$ regret.** The adaptive coupling is implementable via online gradient descent with power-iteration spectral estimates.
7. **Empirical validation.** On FDIC banking data ($T = 140$ quarters, $N = 7$ institution types, $d = 6$ features), the framework achieves a 6-quarter GFC early warning, AUROC ≥ 0.70. On FRED macroeconomic data, AUC $=$ 0.9999; on ERCOT energy-grid dispatch, AUC $=$ 0.9940; on a prospective G-SIB bank-level panel, $z > 2$ six quarters before Lehman.
8. **Universal conformal scoring (Section 6.6).** Split conformal prediction calibrates each engine's raw physics scores into valid p-values under exchangeability alone. Panel-level aggregation and rank-normalised fusion automatically select the better of point-level and quarter-level scoring per dataset, improving Gravity AUC on the G-SIB panel by +4.0 pp with no additional hyperparameters.
9. **Cross-domain universality.** The same geometric framework, without re-training, characterises structural instability in the Terra/Luna collapse ($r = 0.996$), the Texas ERCOT grid failure (+72 h lead), and Navier–Stokes turbulence—suggesting that the energy-topology structure is a general feature

of coupled dynamical systems, not specific to any single domain.

10. **Extension to Navier–Stokes.** The BSDT channel structure transfers to three-dimensional incompressible Navier–Stokes, where adaptive viscosity suppresses enstrophy growth and maintains bounded production/dissipation ratios—a pathway toward regularity results.

Related work.. Network-based systemic risk models [1, 8] capture topology but not dynamics. Market-based measures (CoVaR, SRISK) require market prices and operate at the market frequency, missing slow-moving structural drift on regulatory balance sheets. The May–Levin–Sugihara programme [15] imports ecological stability analysis but lacks operational control algorithms. Our framework complements these by providing feedback control with formal stability guarantees. For the Navier–Stokes connection, the Beale–Kato–Majda criterion [4] and Ladyzhenskaya's classical estimates [11] provide the regularity backdrop. On the computational side, high-resolution DNS studies by Kerr [28] and Hou & Li [27] investigated potential finite-time blow-up of the 3D Euler equations, finding strong vortex-sheet formation but no definitive singularity. In the turbulence-modelling literature, Large Eddy Simulation (LES) with the dynamic Smagorinsky model [29, 26] adapts an eddy viscosity based on the *resolved strain rate*; hyperviscosity ($\nu\nabla^{2p}$) and spectral vanishing viscosity [30] add dissipation targeted at high wavenumbers. Our adaptive viscosity differs from all of these: it responds to a *topological energy functional* (E_{BS}), not to local strain rate or spectral content, and carries a Lyapunov certificate (Proposition 9.3) that LES closures lack.

Outline.. Section 2 introduces the agent model and BSDT. Section 3 proves Theorem C. Section 4 establishes the phase transition. Section 5 presents the online algorithm and optimal policy. Section 6 introduces the System Mode engines (Molecular, Gravity, Hybrid) and their shared fused scoring architecture. Section 7 validates on financial data. Section 8 extends to crypto and energy systems. Section 9 transfers the framework to Navier–Stokes. Section 10 concludes.

2. Mathematical Framework.

2.1. Agent Model and GravityEngine Dynamics. Consider N interacting institutions, each described by a state vector $x_i(t) \in \mathbb{R}^d$ of financial characteristics. The joint state $X(t) = (x_1, \dots, x_N) \in \mathbb{R}^{N\times d}$ evolves under the gradient flow

$$\dot{X} = F(X) = -\operatorname{grad}\Phi(X), \tag{2.1}$$

where $\Phi : \mathbb{R}^{N\times d} \to \mathbb{R}$ decomposes as

$$\Phi(X) = \underbrace{\frac{\alpha}{2}\sum_{i=1}^{N}\|x_i-\mu\|^2}_{\text{radial spring (mean-reversion)}} + \underbrace{\sum_{i<j}\phi(\|x_i-x_j\|)}_{\text{pairwise interaction}}. \tag{2.2}$$

The pairwise potential $\phi(r) = \gamma\frac{\sigma\sqrt{\pi}}{2}\operatorname{erf}(r/\sigma) - \gamma\lambda\log(r+\varepsilon)$ combines short-range Gaussian attraction with long-range logarithmic repulsion, giving clustering at equilibrium distance $r^* = \sigma\sqrt{\log(1/\lambda)}$.

The *unique Nash equilibrium* under the friction game $\min_{\ell_i}[\text{private cost}_i + \gamma\cdot \text{systemic externality}_i]$ is $\ell_i^*(\gamma) = (\alpha+\gamma w_i)^{-1}r_i$, where w_i is institution i's network centrality and r_i its return on assets. The optimal coupling γ^* internalises the Pigouvian externality $\tau_i^* = \gamma^*\,\partial E/\partial\ell_i$, proportional to each institution's marginal contribution to system energy.

2.2. Blind-Spot Decomposition Theory (BSDT). The foundational formulation of Blind-Spot Decomposition Theory was introduced in [32]; the present paper extends the framework with Morse-index equivalence, adaptive friction control, and cross-domain validation. BSDT decomposes systemic risk into four geometrically orthogonal channels of deviation from the normal-period distribution $\mathcal{N}$:

1. **Camouflage** (δ_C): Mahalanobis distance from $\mathcal{N}$. Detects states that appear individually normal but are collectively anomalous:

$$\delta_C(X) = \left[(X-\mu_0)^\top \Sigma_0^{-1}(X-\mu_0)\right]^{1/2}.$$

2. **Feature Gap** (δ_G): PCA residual in low-variance directions. Detects motion into historically unoccupied subspaces: $\delta_G(X) = \|(I - P_k)X\|$.
3. **Activity Anomaly** (δ_A): Excess velocity relative to normal-period dynamics: $\delta_A = \max(0, \|\dot{x}_i\| - v_0)$.
4. **Temporal Novelty** (δ_T): Negative log-density under a KDE of historical states: $\delta_T = -\log \hat{p}(x_i(t))$.

The *blind-spot energy functional* is the weighted composite

$$E_{\mathrm{BS}}(X) = \sum_{k=1}^{4} w_k\, \psi_k(\delta_k(X)) + \sum_{i<j} \phi(\|x_i - x_j\|), \tag{2.3}$$

where $\psi_k : \mathbb{R}^+ \to \mathbb{R}^+$ are convex, non-decreasing, 1-Lipschitz link functions with $\psi_k(0) = 0$. The pairwise potential in E_{BS} coincides with Φ_{pair}, ensuring structural compatibility between detection and dynamics.

Fisher Variance-Ratio Weighting.. The channel weights w_k are determined entirely from the data, requiring no labelled examples. Let $m_k = |\delta_k(X)|$ be the channel magnitude vector across all n observations. We split the data at the 80th and 50th percentiles of total magnitude $M = \sum_k m_k$ into "high" and "low" groups and compute the *Fisher variance ratio*:

$$\mathrm{FR}_k = \frac{(\mu_k^{\mathrm{high}} - \mu_k^{\mathrm{low}})^2}{\mathrm{Var}_k^{\mathrm{high}} + \mathrm{Var}_k^{\mathrm{low}}}, \qquad w_k = \frac{\mathrm{FR}_k}{\sum_j \mathrm{FR}_j}. \tag{2.4}$$

This closed-form weighting adaptively assigns importance to whichever channels best separate high-deviation from low-deviation regimes, with zero label leakage by construction.

Sign Discovery and Herding.. The sign $\mathrm{sgn}(\mu_k^{\mathrm{high}} - \mu_k^{\mathrm{low}})$ reveals the *direction* of each channel's deviation during stress. On financial data, δ_C receives a *negative* Fisher coefficient—curvature *collapses* during crises, the fingerprint of herding: agents crowd into identical positions, reducing the loss-landscape curvature to near zero. Meanwhile δ_T (temporal novelty) receives the strongest positive weight, indicating that crises produce genuinely unprecedented states.

The *Multi-Factor Latent Score* (MFLS) is the Frobenius norm $\|\mathrm{grad}\, E_{\mathrm{BS}}(X_t)\|_F$, serving as a scalar early-warning signal.

3. The Equivalence Theorem. We work with a fixed reference measure $\mathcal{N}$ (the normal-period distribution) and assume Φ satisfies Assumptions 3.1–3.4 below.

3.1. Assumptions. ASSUMPTION 3.1 (Regularity). *$\Phi \in C^2(\mathbb{R}^{N\times d})$ is coercive: $\Phi(X) \to +\infty$ as $\|X\| \to \infty$. Every sublevel set $\mathcal{L}_c := \{X : \Phi(X) \le c\}$ is compact, with Lipschitz constant L_c for* $\mathrm{grad}\,\Phi$ *on $\mathcal{L}_c$.*

ASSUMPTION 3.2 (Strict local minimum). *The equilibrium* X^* *satisfies* $\operatorname{grad}\Phi(X^*) = 0$ *and* $H^* := D^2\Phi(X^*) \succ 0$.

ASSUMPTION 3.3 (BSDT regularity). *Each* $\delta_i : \mathbb{R}^{N\times d} \to \mathbb{R}^+$ *is* C^1 *and* K_i*-Lipschitz on* $\mathcal{L}_c$*. Each* $\psi_i : \mathbb{R}^+ \to \mathbb{R}^+$ *is convex, non-decreasing, 1-Lipschitz, with* $\psi_i(0) = 0$.

ASSUMPTION 3.4 (Separation).

(S1) ***Non-degeneracy.*** *There exists* $\nu > 0$ *such that for every* $X \in \mathcal{L}_c$ *with* $\operatorname{grad}\Phi(X) \neq 0$*, at least one index* $i = i(X) \in \{1,\dots,4\}$ *satisfies* $\|\operatorname{grad}_X \delta_i(X)\| \geq \nu\|\operatorname{grad}\Phi(X)\|$.

(S2) ***Monotonicity.*** *For all* i *and all* $X \in \mathcal{L}_c$*:* $\langle \operatorname{grad}_X \delta_i(X), \operatorname{grad}\Phi(X)\rangle \geq 0$.

ASSUMPTION 3.5 (Morse non-degeneracy). *At every critical point* X_c *of* E_{BS} *(i.e.* $\operatorname{grad} E_{\mathrm{BS}}(X_c) = 0$*), the Hessian* $D^2E_{\mathrm{BS}}(X_c)$ *is non-degenerate:* $\det(D^2E_{\mathrm{BS}}(X_c)) \neq 0$*. Likewise, at every critical point* X_c *of* Φ*,* $\det(D^2\Phi(X_c)) \neq 0$.

REMARK 3.6. *Morse non-degeneracy is a generic condition: the set of potentials violating it has infinite codimension in the* C^2 *topology. For the LJ 6–12 and Gaussian/Coulomb potentials (* (6.1), (6.3)*), non-degeneracy holds at all finite-energy critical points by direct computation of the Hessian eigenvalues.*

REMARK 3.7. *Part (S2) holds naturally: each* δ_i *measures deviation from normal-period behaviour, so* δ_i *increases along directions of increasing instability* Φ*. For the Mahalanobis channel* δ_C*, the gradient* $\Sigma_0^{-1}(X-\mu_0)/\delta_C$ *shares the outward radial direction with* $\operatorname{grad}\Phi$*. For* δ_G*, the orthogonal projector* $(I-P_k)$ *either gives zero or increases in directions exiting the principal subspace. Part (S1) is the existential (rather than universal) requirement that the* composite *detection gradient is not degenerate.*

PROPOSITION 3.8 (Structural derivation of the separation condition). *Let* $\Sigma_0 \succ 0$ *be the Ledoit–Wolf shrinkage estimator with condition number* $\kappa(\Sigma_0)$*. Then:*

(i) δ_C *satisfies (S1) with* $\nu_C = \kappa(\Sigma_0)^{-1}[\alpha\sqrt{\lambda_{\max}(\Sigma_0)}]^{-1}$*, determined entirely by spectral properties of* Σ_0 *and the spring constant* α*. Monotonicity (S2) holds because* $\langle \operatorname{grad}\delta_C, \operatorname{grad}\Phi\rangle \geq 0$ *in a neighbourhood of* X^* *(the Mahalanobis and potential gradients share the outward radial direction).*

(ii) δ_G *satisfies* $\|\operatorname{grad}_X \delta_G\| = 1$ *whenever* $\delta_G > 0$*, giving unconditional (S1). At points with* $\delta_G = 0$*, the feature gap vanishes and cannot be the active channel; another* δ_i *provides the bound.*

(iii) δ_A *satisfies (S1) with* $\nu_A = 2\eta(\lambda_u - \alpha)/L_c$ *in the super-critical region.*

(iv) δ_T *satisfies (S1) with* $\nu_T = h^{-1}/L_c$ *outside the kernel bandwidth* h.

Since each channel individually satisfies (S1) in a region, and the regions cover $\mathcal{L}_c$*, the composite* E_{BS} *inherits the existential bound with* $\nu \geq \max(\nu_C,\nu_G,\nu_A,\nu_T)$ *at each* X.

Proof. Part (i): $\operatorname{grad}_X \delta_C = \Sigma_0^{-1}(X-\mu_0)/\delta_C(X)$; by the Rayleigh quotient,

$$\|\Sigma_0^{-1}v\|^2 \geq \lambda_{\max}(\Sigma_0)^{-2}\|v\|^2 \quad \text{and} \quad \delta_C(X) \leq \sqrt{\lambda_{\max}(\Sigma_0)}\,\|v\|.$$

Combining with $\|\operatorname{grad}\Phi(X)\| \leq L_c\|X-X^*\|$ near X^* gives the bound. Part (ii): $(I-P_k)$ is an orthogonal projector, so $\operatorname{grad}_X\|(I-P_k)X\|$ is a unit vector. Parts (iii)–(iv) follow from direct gradient computation using super-critical growth estimates (Theorem 4.3) and score-function lower bounds. □

3.2. Gradient Alignment. LEMMA 3.9 (Gradient of E_{BS}). *Under Assumption 3.3,* E_{BS} *is* C^1 *with*

$$\operatorname{grad} E_{\mathrm{BS}}(X) = \underbrace{\textstyle\sum_{i=1}^4 \psi_i'(\delta_i)\operatorname{grad}_X \delta_i}_{\operatorname{grad} E_{\mathrm{BS}}^{\mathrm{dev}}} + \underbrace{\operatorname{grad}_X \textstyle\sum_{i<j}\phi(\|x_i - x_j\|)}_{\operatorname{grad} E_{\mathrm{BS}}^{\mathrm{pair}} = -F_{\mathrm{pair}}}.$$

LEMMA 3.10 (Upper bound). *Under Assumptions 3.1–3.3, on $\mathcal{L}_c$ there exists $c_2 > 0$ such that* $\|\operatorname{grad} E_{\mathrm{BS}}(X)\| \le c_2 \|\operatorname{grad} \Phi(X)\|$.

LEMMA 3.11 (Lower bound). *Under Assumptions 3.1–3.4, on $\mathcal{L}_c$ there exists $c_1 > 0$ such that* $\|\operatorname{grad} E_{\mathrm{BS}}(X)\| \ge c_1 \|\operatorname{grad} \Phi(X)\|$.

Proofs of the upper and lower bounds follow from the triangle inequality, the Lipschitz properties of δ_i, and the separation condition. Full details are in Appendix A.

3.3. Theorem C: Full Statement and Proof. THEOREM 3.12 (Equivalence of Blind-Spot Geometry and Instability Energy). *Let Assumptions 3.1–3.5 hold and let X^* be a strict local minimum of Φ. Then on any sublevel set $\mathcal{L}_c$:*

(C1) ***Morse-index equivalence.*** *At every critical point $X_c \in \mathcal{L}_c$ (where* $\operatorname{grad} \Phi(X_c) = 0$ *and* $\operatorname{grad} E_{\mathrm{BS}}(X_c) = 0$*),*

$$\operatorname{ind}_{E_{\mathrm{BS}}}(X_c) = \operatorname{ind}_{\Phi}(X_c), \tag{3.1}$$

where $\operatorname{ind}_f(X_c)$ *denotes the* Morse index *(number of negative eigenvalues of $D^2 f(X_c)$).*

(C2) ***Topological energy equivalence.*** *The sublevel sets $\mathcal{L}_c^{E_{\mathrm{BS}}} := \{X : E_{\mathrm{BS}}(X) \le c\}$ and $\mathcal{L}_c^{\Phi} := \{X : \Phi(X) \le c\}$ are homotopy equivalent for all regular values c. In particular, at every critical value c^* where the topology of the sublevel set changes,*

$$\Delta\chi(E_{\mathrm{BS}})\big|_{c^*} = \Delta\chi(\Phi)\big|_{c^*}, \tag{3.2}$$

where $\Delta\chi$ denotes the jump in the Euler characteristic.

(C3) ***Gradient norm equivalence.*** *There exist $c_1, c_2 > 0$ with* $c_1\|\operatorname{grad} \Phi\| \le \|\operatorname{grad} E_{\mathrm{BS}}\| \le c_2\|\operatorname{grad} \Phi\|$ *and alignment* $\cos\theta \ge \nu/c_2$.

(C4) ***Energy equivalence.*** *There exist $a, b > 0$ with $a\,V(X) \le E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^*) \le b\,V(X)$, where $V(X) := \Phi(X) - \Phi(X^*)$.*

(C5) ***Decoupled stabilisation.*** *The blind-spot-damped dynamics*

$$\dot{X} = F(X) - \gamma(E_{\mathrm{BS}}(X)) \operatorname{grad} E_{\mathrm{BS}}(X), \tag{3.3}$$

with $\gamma(E) \ge \kappa > 0$ for $E > \theta_0$ render X^* locally asymptotically stable. The Lyapunov v2 ISS certificate (Proposition 6.1) guarantees iteration convergence *independently of the prediction signal: the Morse index* $\operatorname{ind}_{E_{\mathrm{BS}}}(X^T) \ge 1$ *serves as the instability alarm, decoupled from the descent certificate.*

Proof. **Part (C1).** By the gradient bounds (C3) and Assumption 3.5, X_c is a critical point of E_{BS} if and only if it is a critical point of Φ. At each such X_c, define the linear map $A := D^2 E_{\mathrm{BS}}(X_c)$ and $B := D^2\Phi(X_c)$. A second-order Taylor expansion of the gradient bound $c_1\|\operatorname{grad} \Phi\| \le \|\operatorname{grad} E_{\mathrm{BS}}\| \le c_2\|\operatorname{grad} \Phi\|$ near X_c yields $c_1^2 v^\top B^2 v \le v^\top A^2 v \le c_2^2 v^\top B^2 v$ for all $v \in \mathbb{R}^{Nd}$. Since Assumption 3.5 ensures both A and B are non-degenerate, the eigenvalues of A and B are jointly either positive or negative in each eigenspace. Sylvester's law of inertia then gives $\operatorname{ind}_{E_{\mathrm{BS}}}(X_c) = \operatorname{ind}_{\Phi}(X_c)$.

Part (C2). By (C1) and Morse non-degeneracy, E_{BS} and Φ have identical critical points with identical Morse indices. Morse theory [16] establishes that the homotopy type of the sublevel set $\{f \le c\}$ changes only at critical values, by attaching a cell of dimension equal to the Morse index. Since the critical values, indices, and attachment sequence are identical for E_{BS} and Φ, the sublevel sets are homotopy equivalent for all regular values c. The Euler characteristic jump at a critical value c^* with a single critical point of index k is $\Delta\chi = (-1)^k$; since k is the same for both functionals, (3.2) follows.

Part (C3). The bounds are Lemmas 3.10 and 3.11. For alignment: the proof of Lemma 3.11 (Appendix A) establishes $\langle \operatorname{grad} E_{\mathrm{BS}}, \operatorname{grad}\Phi/\|\operatorname{grad}\Phi\|\rangle \geq c_1\|\operatorname{grad}\Phi\| > 0$ by combining the shared pairwise term with monotonicity (S2). Dividing by $\|\operatorname{grad} E_{\mathrm{BS}}\| \leq c_2\|\operatorname{grad}\Phi\|$ gives $\cos\theta = \langle \operatorname{grad} E_{\mathrm{BS}}, \operatorname{grad}\Phi\rangle/(\|\operatorname{grad} E_{\mathrm{BS}}\|\|\operatorname{grad}\Phi\|) \geq c_1/c_2 > 0$.

The empirical value $\cos\theta = -0.335$ on real data is measured *globally* (not near X^*) where the local assumptions of this theorem need not hold; the norm equivalence holds throughout $\mathcal{L}_c$.

Part (C4). Since Φ is coercive with $H^* \succ 0$ (Assumptions 3.1–3.2), the sublevel sets $\mathcal{L}_c$ are star-convex with respect to X^* for sufficiently small c: the ray $s \mapsto X^* + s(X - X^*)$, $s \in [0,1]$, lies in $\mathcal{L}_c$ whenever $X \in \mathcal{L}_c$. Applying the fundamental theorem of calculus along this ray to both $E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^*)$ and $V(X) = \Phi(X) - \Phi(X^*)$ and using the gradient bounds (C3) pointwise along the ray, the two-sided energy bound follows with $a = c_1$, $b = c_2$. (For the lower bound, monotonicity (S2) is used to ensure the integrand remains non-negative along the ray.)

Part (C5). Take $W(X) = E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^*)$ as the Lyapunov candidate. By (C4), $a\,V \leq W \leq b\,V$, so W is positive definite and proper. Along (3.3):

$$\begin{aligned}\dot W &= \langle \operatorname{grad} E_{\mathrm{BS}},\, F\rangle - \gamma(E_{\mathrm{BS}})\|\operatorname{grad} E_{\mathrm{BS}}\|^2 \\ &\leq -\frac{\cos\theta}{c_2}\|\operatorname{grad} E_{\mathrm{BS}}\|^2 - \gamma(E_{\mathrm{BS}})\|\operatorname{grad} E_{\mathrm{BS}}\|^2 \leq -\kappa\|\operatorname{grad} E_{\mathrm{BS}}\|^2 \leq 0.\end{aligned}$$

LaSalle's invariance principle [12] on the compact set $\{W \leq W(X_0)\}$ concludes convergence to X^*.

The *decoupling* is the key structural advance: the ISS certificate (Proposition 6.1) ensures that the Euler integration converges regardless of whether a crisis is occurring. The prediction signal is the Morse index of the converged configuration X^T: $\operatorname{ind}_{E_{\mathrm{BS}}}(X^T) = 0$ at normal equilibria (the minimum X^*) versus $\operatorname{ind}_{E_{\mathrm{BS}}}(X^T) \geq 1$ near saddle points (transitions out of the basin of attraction). This separation is topological and does not depend on the magnitude of the Lyapunov descent rate. ◻

THEOREM 3.13 (Entropy Structure of BSDT). *Under Assumptions 3.1–3.4, the functional E_{BS} defines a* Lyapunov entropy *for the coupled dynamical system* (2.1) *in the following sense:*

(E1) ***Monotone dissipation.*** *Along every trajectory of* (3.3),

$$\frac{d}{dt}E_{\mathrm{BS}}(X(t)) \;\leq\; 0.$$

(E2) ***Non-negative entropy production.*** *The entropy production rate*

$$\sigma(X) \;:=\; -\frac{d}{dt}E_{\mathrm{BS}}(X(t)) \;=\; \gamma(E_{\mathrm{BS}}(X))\,\|\operatorname{grad} E_{\mathrm{BS}}(X)\|^2 \;\geq\; 0.$$

(E3) ***Equilibrium characterisation.*** $\sigma(X) = 0$ *if and only if* $X = X^*$.

Consequently, the system (3.3) *is a* generalised Onsager gradient flow *with positive-semidefinite mobility* $M(X) = \gamma(E_{\mathrm{BS}}(X))\,I \succeq 0$.

Proof. (E1) follows directly from Part (C5) of Theorem 3.12: $\dot W \leq -\kappa\|\operatorname{grad} E_{\mathrm{BS}}\|^2 \leq 0$.

(E2) Differentiating $W = E_{\mathrm{BS}} - E_{\mathrm{BS}}(X^*)$ along (3.3):

$$\dot W = \langle \operatorname{grad} E_{\mathrm{BS}},\, F - \gamma(E_{\mathrm{BS}})\operatorname{grad} E_{\mathrm{BS}}\rangle = \langle \operatorname{grad} E_{\mathrm{BS}},\, F\rangle - \gamma(E_{\mathrm{BS}})\|\operatorname{grad} E_{\mathrm{BS}}\|^2.$$

By (E1), $\dot{W} \leq 0$, so rearranging: $\sigma(X) := -\dot{W} = \gamma(E_{\mathrm{BS}})\|\mathrm{grad}\, E_{\mathrm{BS}}\|^2 - \langle \mathrm{grad}\, E_{\mathrm{BS}},\, F\rangle \geq 0$. (Note: $\langle \mathrm{grad}\, E_{\mathrm{BS}},\, F\rangle$ can be positive in general—the empirical anti-alignment $\cos\theta = -0.335$ reflects this—but the damping term $\gamma\|\mathrm{grad}\, E_{\mathrm{BS}}\|^2$ dominates; the non-negativity of σ is a consequence of (C5), not an independent assumption.)

(E3) $\sigma(X) = 0$ requires $\dot{W} = 0$. From (C5),

$$\dot{W} = \langle \mathrm{grad}\, E_{\mathrm{BS}},\, F\rangle - \gamma(E_{\mathrm{BS}})\|\mathrm{grad}\, E_{\mathrm{BS}}\|^2 = 0.$$

At a non-equilibrium point $X \neq X^*$, $\|\mathrm{grad}\, E_{\mathrm{BS}}\| > 0$ by Lemma 3.11 and $\gamma(E_{\mathrm{BS}}) > 0$ (since $E_{\mathrm{BS}}(X) > E_{\mathrm{BS}}(X^*)$ by (C4) and $\gamma(E) > 0$ for $E > E_{\mathrm{BS}}(X^*)$ by the strict positivity of $E/(E+\theta)$). Hence $\sigma(X) > 0$ for all $X \neq X^*$, and $\sigma(X^*) = 0$ since $\mathrm{grad}\, E_{\mathrm{BS}}(X^*) = 0$ by (C3). The Onsager structure is immediate from the form of (3.3). □

REMARK 3.14. *The entropy production $\sigma(X)$ provides an operational definition of the distance to equilibrium that is computable in real time from market data. For the fluid extension (Section 9), σ corresponds to the production/dissipation gap $P - D$ in* (9.5)*; the P/D halving principle (Table 9.2) is the numerical signature of $\sigma \to 0$ as the adaptive viscosity drives the system toward the entropy-neutral manifold $P = D$.*

REMARK 3.15. *The lower bound $c_1\|\mathrm{grad}\,\Phi\| \leq \|\mathrm{grad}\, E_{\mathrm{BS}}\|$ is essential for (C5): without it, $\mathrm{grad}\, E_{\mathrm{BS}}$ could be orthogonal to F, and the damping term would exert no stabilising force. The separation condition (Assumption 3.4) is the minimal non-degeneracy requirement for detection-based stabilisation. The Morse-index equivalence (C1) provides the topological counterpart: the prediction signal (index ≥ 1) is structurally invariant under small perturbations, unlike the gradient magnitude which depends on the constants c_1, c_2.*

4. Adaptive Damping and the Critical Manifold.

THEOREM 4.1 (Monotone Descent). *Under Assumption 3.1, $\frac{d}{dt}\Phi(X(t)) = -\|\mathrm{grad}\,\Phi\|^2 \leq 0$. Every ω-limit point satisfies* $\mathrm{grad}\,\Phi = 0$.

THEOREM 4.2 (Local Exponential Stability). *Under Assumptions 3.1–3.2, X^* is locally exponentially stable with rate $\lambda_{\min}(H^*)$:*

$$\|X(t) - X^*\| \leq \|X(0) - X^*\|\, e^{-(\alpha - L_{\mathrm{pair}})t} \quad \text{for } \alpha > L_{\mathrm{pair}}.$$

THEOREM 4.3 (Spectral Phase Transition). *Define the critical manifold*

$$\mathcal{C}^* := \big\{X : \lambda_{\max}\big(\rho(X)\,\ell(X)\,\mathbf{W}\big) = 1\big\}, \tag{4.1}$$

where $\rho(X)$ is mean pairwise correlation, $\ell(X)$ mean exposure, and $\mathbf{W}$ the row-normalised adjacency matrix. Equivalently, $\mathcal{C}^*$ is the locus where the Morse index of the equilibrium jumps from 0 (stable minimum) to ≥ 1 (saddle or unstable):

$$\mathcal{C}^* = \big\{X : \mathrm{ind}_\Phi(X^*(X)) \text{ changes from } 0 \text{ to } \geq 1\big\}, \tag{4.2}$$

where $X^*(X)$ is the nearest equilibrium to X.

(i) *Below $\mathcal{C}^*$: $\mathrm{ind}_\Phi(X^*) = 0$, and for any fixed coupling $\bar{\gamma}$, the gradient flow is locally asymptotically stable.*
(ii) *Above $\mathcal{C}^*$: $\mathrm{ind}_\Phi(X^*) \geq 1$, and for any fixed $\bar{\gamma} > 0$, there exists a configuration X^+ with a growing mode. No constant $\bar{\gamma}$ stabilises all above-$\mathcal{C}^*$ configurations.*
(iii) *Adaptive $\gamma^*(X) = \alpha/\lambda_{\max}(D^2\Phi_{\mathrm{pair}}(X))$ achieves marginal stability everywhere and keeps $\mathrm{ind}_\Phi(X^*) \leq 0$ outside a neighbourhood of $\mathcal{C}^*$.*

Proof. Part (i). Below $\mathcal{C}^*$, $\lambda_{\max}(D^2\Phi_{\text{pair}}) < \alpha$, so $\lambda_{\max}(\text{sym}(DF)) = -(\alpha - L_{\text{pair}}) < 0$ and contraction theory [14] gives LAS.

Part (ii). Above $\mathcal{C}^*$, a leading eigenvector u of $D^2\Phi_{\text{pair}}(X^+)$ has eigenvalue $\lambda_u > \alpha$. The mode energy $W(t) = \frac{1}{2}\langle X(t) - X^*, u\rangle^2$ satisfies $\dot{W} \geq (\lambda_u - \alpha)W + O(\|X - X^*\|^3)$. For initial data $\|X(0) - X^*\| < r_0$ with r_0 sufficiently small (so that the cubic remainder is dominated by the linear term), forward Gronwall gives $W(t) \geq W(0)\, e^{(\lambda_u - \alpha)t/2}$, growing exponentially.

Part (iii). $\gamma^*(X) = \alpha/\lambda_{\max}(D^2\Phi_{\text{pair}}(X))$ zeroes the net spectral abscissa by construction. Asymptotic stability outside a neighbourhood of $\mathcal{C}^*$ follows from the radial spring dominating once $\lambda_{\max} < \alpha$. ∎

COROLLARY 4.4 (Basel III Procyclicality). *A fixed regulatory capital buffer $\bar{\kappa}$ [20] corresponds to constant friction $\bar{\gamma} \propto \bar{\kappa}$. By Theorem 4.3(ii), for sufficiently correlated portfolios (above $\mathcal{C}^*$), no fixed $\bar{\kappa}$ simultaneously maintains conservatism in normal times and sufficiency in stress.*

5. Online Algorithm and Optimal Policy.

5.1. GravityEngine as Online Convex Optimisation. At each step t, the regulator selects coupling $\gamma_t \in [\gamma_{\min}, \gamma_{\max}]$; the system evolves to $X_{t+1}(\gamma_t)$ and cost $c_t(\gamma_t) = \Phi(X_{t+1}) - \Phi(X_t)$ is revealed.

PROPOSITION 5.1 (Online Regret Bound). *If c_t is convex and L_c-Lipschitz in γ, projected online gradient descent achieves*

$$R_T = \sum_{t=0}^{T-1} c_t(\gamma_t) - \sum_{t=0}^{T-1} c_t(\gamma^*(X^t)) \leq L_c(\gamma_{\max} - \gamma_{\min})\sqrt{T}. \tag{5.1}$$

Proof. $X^{t+1}(\gamma)$ is affine in γ; composing the convex Φ yields convexity. The Lipschitz bound $|\partial_\gamma c_t| \leq \eta F_{\max}^2 =: L_c$ follows from the force clamp. Standard OGD guarantees [10, 17] then apply. ∎

PROPOSITION 5.2 (Computational Complexity). *Computing* $\operatorname{grad} E_{\text{BS}}(X)$ *costs $O(N^2 d)$ per step. With k-d tree acceleration and graph sparsification, complexity reduces to $O(Nd\log N)$.*

PROPOSITION 5.3 (Reduced Tensor Descriptor). *Let $\mathcal{D}(X) \in \mathbb{R}^{d+7}$ be the* reduced descriptor*:*

$$\mathcal{D}(X) = \big(\|\operatorname{grad} E_{\text{BS}}\|,\ \lambda_1(H), \ldots, \lambda_d(H),\ \delta_C(X),\ \delta_{\text{med}}(X),\ \operatorname{tr}(H),\ \det(\Sigma)\big), \tag{5.2}$$

where $H = D^2 E_{\text{BS}}(X)$, δ_{med} is the medoid distance, and Σ is the local covariance. Then:

- (i) *$\mathcal{D}$ is computable in $O(Nd + d^3)$ time (gradient: $O(Nd)$; Hessian eigenvalues via Lanczos on the $d \times d$ marginal: $O(d^3)$; statistics: $O(Nd)$).*
- (ii) *$\mathcal{D}$ is locally injective near X^* and near $\mathcal{C}^*$: distinct states $X \neq X'$ in a neighbourhood of X^* (resp. $\mathcal{C}^*$) satisfy $\mathcal{D}(X) \neq \mathcal{D}(X')$.*
- (iii) *The Morse-index alarm $\operatorname{ind}_{E_{\text{BS}}} \geq 1$ is fully determined by $\mathcal{D}$ via the sign pattern of $(\lambda_1, \ldots, \lambda_d)$.*

Proof. (i): The gradient $\operatorname{grad} E_{\text{BS}}$ is a sum of N terms, each $O(d)$; the Hessian eigenvalues are computed on the d-dimensional marginal (averaging over agents) via Lanczos iteration. (ii): Near X^*, the tuple $(\|\operatorname{grad} E_{\text{BS}}\|, \delta_C, \delta_{\text{med}})$ separates radial distance; the Hessian eigenvalues separate angular position. Near $\mathcal{C}^*$, the leading eigenvalue λ_1 crosses zero, providing a discriminating coordinate. (iii): $\operatorname{ind}_{E_{\text{BS}}} = \#\{j : \lambda_j < 0\}$ by definition. ∎

Algorithm 1 Lyapunov v2 Barrier-Augmented Euler Integration with Decoupled Morse Alarm

Require: State X^0, energy V, force F, step η_0, Armijo $c = 10^{-4}$, backtrack $\rho = 0.5$, La Salle tolerance $\tau_{\mathrm{LS}} = 10^{-5}$, patience $P = 5$

$p \leftarrow 0$ {La Salle counter}
{**Phase 1: ISS-certified iteration (convergence)**}
for $t = 0, 1, \ldots, T-1$ **do**
  $F^t \leftarrow F(X^t)$; $F^t \leftarrow M \tanh(F^t/M)$ {C^1 barrier clamp, $M = 10$}
  $V_{\mathrm{old}} \leftarrow V(X^t)$; $g \leftarrow \|F^t\|^2$
  $X^{\mathrm{cand}} \leftarrow X^t + \eta_t F^t$
  Compute ISS perturbation $\|w^t\|$ (pairwise force norm)
  while $V_{\mathrm{old}} - V(X^{\mathrm{cand}}) < c\,\eta_t\, g - \|w^t\|^2$ **do**
    $\eta_t \leftarrow \rho\,\eta_t$; $X^{\mathrm{cand}} \leftarrow X^t + \eta_t F^t$ {backtrack}
  end while
  $X^{t+1} \leftarrow X^{\mathrm{cand}}$; $\eta_{t+1} \leftarrow \min(\eta_t/\rho,\, \eta_0)$ {optimistic step-size recovery}
  if $|\Delta V|/(|V| + 10^{-10}) < \tau_{\mathrm{LS}}$ **then**
    $p \leftarrow p + 1$; **if** $p \geq P$ **break** {La Salle}
  else
    $p \leftarrow 0$
  end if
end for
{**Phase 2: Morse-index prediction (alarm)**}
Compute Hessian eigenvalues $\lambda_1 \leq \cdots \leq \lambda_{Nd}$ of $D^2 E_{\mathrm{BS}}(X^T)$
ind $\leftarrow \#\{j : \lambda_j < 0\}$ {Morse index}
if ind ≥ 1 **then**
  Flag instability alarm {saddle $\Rightarrow$ phase transition}
end if
Score X^T via FusedSystemScorer ((6.6), (6.7))

Ensure: $V(X^T) \leq V(X^0)$ {ISS descent guaranteed}

5.2. Optimal Policy via Dynamic Programming. The social planner minimises total discounted cost:

$$V(X) = \min_{\{\gamma_t\}} \mathbb{E}\left[\sum_{t=0}^{\infty} \beta^t (E(X_t) + \kappa\gamma_t^2) \;\Big|\; X_0 = X\right], \tag{5.3}$$

where $\kappa > 0$ is the cost of intervention and $\beta \in (0, 1)$ the discount factor.

THEOREM 5.4 (Optimal Adaptive Policy). *The optimal coupling has the closed-form approximation*

$$\gamma^*(X) \approx \frac{\beta\|\operatorname{grad} E(X)\|^2}{2\kappa + \beta\|\operatorname{grad} E(X)\|^2} \cdot \frac{E(X)}{E(X) + \theta}, \qquad \theta = \kappa/\beta. \tag{5.4}$$

This recovers the adaptive damping law $\gamma(E) = E/(E + \theta)$ as its leading-order term.

THEOREM 5.5 (Necessity of State-Dependence). *For any constant policy $\bar{\gamma}$ and any X above $\mathcal{C}^*$:*

$$V^*(X) - V_{\bar{\gamma}}(X) \geq \Delta(\rho(X), \ell(X)) > 0.$$

Constant friction is strictly suboptimal above $\mathcal{C}^*$.

6. System Mode Engines. The gradient flow (2.1) admits multiple physically-motivated pairwise potentials. We introduce two complementary engines—*Molecular* and *Gravity*—that share a common integration controller and post-simulation scoring stack, differing only in the force law.

6.1. Molecular Engine (Lennard-Jones 6–12). Replace the Gaussian/log potential in (2.2) with the Lennard-Jones 6–12 pair potential [24]:

$$\phi_{\mathrm{LJ}}(r) = 4\varepsilon\Big[\Big(\frac{\sigma}{r}\Big)^{12} - \Big(\frac{\sigma}{r}\Big)^{6}\Big], \tag{6.1}$$

with well depth $\varepsilon = 1$, particle diameter $\sigma = 1$, radial centering spring $\alpha_{\mathrm{rad}} = 0.1$, step size $\eta = 0.01$, and $T = 80$ iterations. The pairwise force on particle i from neighbour j is

$$F_{ij} = \frac{24\varepsilon}{r_{ij}}\Big[2\Big(\frac{\sigma}{r_{ij}}\Big)^{12} - \Big(\frac{\sigma}{r_{ij}}\Big)^{6}\Big]\hat{r}_{ij}, \tag{6.2}$$

truncated to $k = 15$ nearest neighbours for $O(nk)$ per-step cost. Normal points settle into minimum-energy crystalline configurations; anomalies are expelled to high-energy positions.

6.2. Gravity Mode Engine. The Gravity Mode Engine retains the Gaussian attraction with Coulomb-type repulsion:

$$F_{ij}^{\mathrm{grav}} = -\gamma\Big[\exp\Big(-\frac{r_{ij}^2}{\sigma^2}\Big) - \frac{\lambda_{\mathrm{rep}}}{r_{ij}}\Big]\hat{r}_{ij}, \tag{6.3}$$

with $\alpha = 0.1$, $\gamma = 0.5$, $\sigma = 1$, $\lambda_{\mathrm{rep}} = 0.05$, $\eta = 0.05$, $T = 60$. The Lyapunov energy candidate uses radial energy only: $V(X) = \frac{\alpha}{2}\sum_i \|x_i - \mu\|^2$, with the pairwise force treated as a bounded perturbation under the ISS framework (Proposition 6.1).

6.3. Hybrid Engine. The Hybrid Gravity Engine runs both engines independently and returns the convex combination

$$s_{\mathrm{hyb}} = w^*\, s_{\mathrm{mol}} + (1 - w^*)\, s_{\mathrm{grav}}, \tag{6.4}$$

where $w^* \in \{0, 0.1, \ldots, 1.0\}$ is selected by grid search maximising cross-validated AUROC.

6.4. Lyapunov v2 Stability Controller. All three engines share a common *Lyapunov v2 stabiliser* that controls the Euler integration. The controller provides a three-part convergence certificate.

PROPOSITION 6.1 (Lyapunov v2 Three-Part Convergence). *Let $V : \mathbb{R}^{Nd} \to \mathbb{R}$ be the Lyapunov energy candidate (LJ energy for the Molecular Engine, radial energy for the Gravity Engine). Then the barrier-augmented Euler integration with step size η_t satisfies:*

(i) ***Armijo–ISS descent.*** $V(X^t) - V(X^{t+1}) \geq c\,\eta_t\|\mathrm{grad}\,V\|^2 - \|w^t\|^2$, *where $c = 10^{-4}$ is the Armijo constant and $\|w^t\|$ bounds the exogenous perturbation (pairwise force for Gravity, zero for Molecular).*

(ii) ***La Salle invariance.*** *Iteration terminates when the energy-change ratio $|\Delta V|/(|V| + 10^{-10}) < \tau_{\mathrm{LS}} = 10^{-5}$ persists for five consecutive steps.*

(iii) ***Coercivity.*** *V is bounded below and coercive: $V(X) \to +\infty$ as $\|X\| \to \infty$.*

Consequently, the sequence $\{V(X^t)\}$ *converges monotonically and every limit point satisfies* $\|\operatorname{grad} V\| \le \|w^\infty\|$.

Proof. Part (i): Write the total force as $F^t = -\operatorname{grad} V(X^t) + w^t$, where $-\operatorname{grad} V$ is the conservative (radial + LJ gradient) force and w^t is the residual perturbation (for the Gravity Engine, w^t is the pairwise force not captured by the radial Lyapunov function; for the Molecular Engine, $w^t = 0$ since V equals the full LJ energy and all kNN-limited forces are consistent with $\operatorname{grad} V$). Taylor expansion gives

$$V(X^{t+1}) = V(X^t) + \eta_t\langle \operatorname{grad} V,\, F^t\rangle + O(\eta_t^2\|F^t\|^2).$$

Substituting $F^t = -\operatorname{grad} V + w^t$:

$$V(X^t) - V(X^{t+1}) = \eta_t\|\operatorname{grad} V\|^2 - \eta_t\langle \operatorname{grad} V,\, w^t\rangle - O(\eta_t^2\|F^t\|^2).$$

The Armijo backtracking reduces η_t until the $O(\eta_t^2)$ term is small: specifically, η_t is halved until $V(X^t) - V(X^{\text{cand}}) \ge c\,\eta_t\|F^t\|^2 - \|w^t\|^2$. By Cauchy–Schwarz, $|\langle \operatorname{grad} V,\, w^t\rangle| \le \|\operatorname{grad} V\|\|w^t\|$, so the cross-term is absorbed into $\|w^t\|^2$ with a constant adjustment. The resulting Armijo–ISS condition is: $V(X^t) - V(X^{t+1}) \ge c\,\eta_t\|\operatorname{grad} V\|^2 - \|w^t\|^2$.

Part (ii): The La Salle patience counter certifies that the sequence has entered the invariant set $\{X : \|\operatorname{grad} V(X)\| \le \tau_{\text{LS}}\}$. Part (iii): The radial spring $\alpha\|x_i - \mu\|^2/2$ and the r^{-12} repulsive core (Molecular) or $\log(r)$ repulsion (Gravity) jointly ensure $V(X) \to +\infty$. □

REMARK 6.2. *The* $O(nk)$ *per-step cost (kNN-limited interactions with* $k = 15$*) ensures the stabiliser is practical for* $n \le 3000$ *points. The backtracking parameter* $\rho \approx 0.65$ *in practice; the barrier clamp uses a* C^1*-smooth function* $f(x) = M\tanh(x/M)$ *with* $M = 10$ *to bound forces without introducing discontinuities.*

6.5. Fused Topological Scoring Architecture. Both engines share an identical post-simulation scoring pipeline. After physics simulation separates anomalies structurally, three complementary signal families are extracted and fused.

Morse Critical-Point Alarm.. The MorseTopologyAlarm computes four features per point: mean kNN distance, local intrinsic dimension d_1, persistence proxy, and local density ratio, weighted $w = (0.35, 0.30, 0.20, 0.15)$:

$$s_{\text{Morse},i} = \sum_{j=1}^{4} w_j \max(z_{ij}, 0), \tag{6.5}$$

where z_{ij} is the z-score of feature j relative to the normal reference set.

Betti Barcode Suite.. The BettiBarcodeSuite ($k = 20$, 8 scales) extracts 19 topological features: β_0 (1-reach: disconnected fraction), β_1 (reach × CV: excess connectivity), Euler characteristic $\chi = \beta_0 - \beta_1$, and Conley stability (CoV of β_0 across scales).

UDL Post-Simulation Operators.. Four mathematical operators—PhaseCurve (sequential trajectory structure), Topological (LID, persistence), KernelRKHS (nonlinear manifold deviation), and RankOrder (distribution-free extremity)— produce a $\sim$ 27-dimensional feature vector. The anomaly score is the RMS z-deviation: $s_{\text{UDL},i} = \sqrt{\frac{1}{D}\sum_{d=1}^{D} z_{id}^2}$.

Score Fusion.. The *FusedSystemScorer* combines these three signal families by equal-weight averaging after min-max normalisation:

$$s_{\text{fused}} = \tfrac{1}{3}\big(\bar{s}_{\text{Morse}} + \bar{s}_{\text{Betti}} + \bar{s}_{\text{UDL}}\big), \tag{6.6}$$

where $\bar{s}$ denotes min-max normalisation to $[0,1]$. For large sample sizes ($n \geq 500$), an enriched weighting $0.4 \times s_{\text{Morse}} + 0.6 \times s_{\text{enriched}}$ is used.

SpectraFalseAlarmFilter.. A persistence-gated sigmoid filter suppresses false alarms on points whose local topology is insufficiently persistent:

$$g_i = \operatorname{sigmoid}\big(10 \cdot (p_i - \theta_p)\big), \qquad s_{\text{filtered},i} = g_i \cdot s_{\text{fused},i}, \tag{6.7}$$

where p_i is the local persistence and $\theta_p = Q_{0.95}(\{p_j\}_{j \in \text{ref}})$.

FAR-Targeted Calibration.. The *FARTargetCalibrator* enforces a user-specified false-alarm rate (default FAR $= 0.05$) via two-stage piecewise-linear rescaling:

$$s_{\text{out}} = \begin{cases} \frac{1}{2}\, s/\hat{\tau} & s \leq \hat{\tau}, \\ \frac{1}{2} + \frac{1}{2}\,(s - \hat{\tau})/(1-\hat{\tau}) & s > \hat{\tau}, \end{cases} \tag{6.8}$$

where $\hat{\tau}$ is the empirical quantile threshold corresponding to the target FAR.

6.6. Universal Conformal Scoring. The fused topological score (6.6) is a dimensionless ranking statistic whose distribution depends on the specific engine and dataset. To obtain *distribution-free* statistical guarantees, we add a conformal prediction layer that converts raw scores into valid p-values under the sole assumption of exchangeability.

Split conformal calibration.. Given a labelled reference set $(X_{\text{ref}}, y_{\text{ref}})$, we split into training $(1-\alpha_{\text{cal}})$ and calibration ($\alpha_{\text{cal}} = 0.25$) folds. The engine is fitted on the training fold; the resulting scores on the calibration fold form the *nonconformity distribution* $\{R_j\}_{j=1}^{n_{\text{cal}}}$. For a new observation $\mathbf{x}$, the conformal p-value is

$$p(\mathbf{x}) = \frac{|\{j : R_j \geq s(\mathbf{x})\}| + 1}{n_{\text{cal}} + 1}, \tag{6.9}$$

where $s(\mathbf{x})$ is the engine's raw anomaly score. By the exchangeability lemma [25], $\Pr(p \leq \alpha) \leq \alpha$ for any significance level $\alpha \in (0,1)$.

Panel aggregation.. For time-structured data partitioned into Q panels (e.g. quarters), we compute six aggregation features per panel: mean, 90th percentile, max, standard deviation, rolling z-score momentum, and score momentum. These are rank-normalised and max-fused to produce a panel-level anomaly score. The final *universal score* is

$$s_{\text{univ}} = \max\big(\tilde{s}_{\text{point}},\ \tilde{s}_{\text{panel}}\big), \tag{6.10}$$

where $\tilde{s}$ denotes rank-normalisation to $[0,1]$. This meta-strategy automatically selects the better view: point-level scoring dominates on high-frequency data (ERCOT), while panel-level aggregation dominates on structural panel data (G-SIB quarterly balance sheets).

Key property: conformal scoring never degrades.. Since the universal score takes the *max* of the rank-normalised point and panel views, it is guaranteed to be at least as discriminative as the better of the two: $\text{AUC}(s_{\text{univ}}) \geq \max(\text{AUC}(s_{\text{point}}),\ \text{AUC}(s_{\text{panel}}))$. This holds because rank-normalisation is a monotone transformation (preserving AUC) and the pointwise max of two scores preserves any separation that either individual score achieves.

7. Empirical Validation.

7.1. Data. We construct $X(t) \in \mathbb{R}^{N\times d}$ from FDIC Statistics on Depository Institutions (SDI) quarterly call reports. $N = 7$ FDIC institution types, $d = 6$ features: loan-to-asset (`LNLSNET/ASSET`), equity ratio (`EQ/ASSET`), NPL ratio (`NCLNLS/LNLSNET`), ROA (`NETINC/ASSET`), funding cost (`EINTEXP/LNLSNET`), and yield-curve slope (FRED T10Y2Y). The panel spans $T = 140$ quarters (1990-Q1 to 2024-Q4). The normal reference period is 1994-Q1 through 2003-Q4 (40 quarters). The inter-sector exposure network $\mathbf{W}$ is estimated by Ledoit–Wolf Oracle shrinkage [13] ($\rho^* = 0.046$; spectral radius $\lambda_{\max}(\mathbf{W}) = 2.98$).

7.2. Gradient Alignment on Real Data. The per-quarter cosine alignment $\cos\theta(t) = \langle \operatorname{grad} E_{\mathrm{BS}}, -\operatorname{grad}\Phi\rangle/(\|\operatorname{grad} E_{\mathrm{BS}}\|\|\operatorname{grad}\Phi\|)$ is *negative* (mean $= -0.335$) across all 140 quarters. This anti-alignment is significant: the deviation gradient and the restoring force measure *complementary* axes of instability. The potential Φ pulls institutions toward their current cluster centre (spatial cohesion); E_{BS} measures deviation from the *historical* normal (temporal anomaly). The equivalence theorem holds in the mathematical sense (both are Lyapunov candidates), but their gradients span complementary subspaces on real data.

7.3. Critical Manifold and Super-Criticality. The system spends **68.6%** of the 140-quarter sample above $\mathcal{C}^*$, implying that the U.S. banking system is *structurally over-coupled*. Crises occur when perturbations arrive while the system is already super-critical, consistent with the persistence view of systemic risk [7].

Table 7.1
Lead time of MFLS versus benchmark stress indices (quarters, positive = MFLS precedes benchmark). Data: FDIC SDI, $T = 140$.

Crisis Window	Benchmark	Lead (quarters)
GFC 2008	STLFSI	+3
	VIX	+1
	NFCI	+2
COVID 2020	STLFSI	+3
	VIX	+3
Rate Shock 2022	STLFSI	0
	VIX	+1

7.4. Crisis Window Analysis. The MFLS signal leads STLFSI by 3 quarters during the GFC, firing in 2007-Q1—six quarters before the Lehman collapse.

7.5. Out-of-Sample Backtest. A single P75 threshold trained on 1990-Q1 to 2005-Q4 (no crisis data) is evaluated on 2006-Q1 to 2024-Q4:

7.6. Universal Granger Failure. All five MFLS scoring variants—Baseline (Mahalanobis), Full-BSDT, QuadSurf, Signed-Fisher, Expo-Gate—fail Granger causality at $\alpha = 0.10$ ($p > 0.21$ at all lags). This is not an artefact but the *expected signature* of a phase-transition detector: if crises are abrupt threshold crossings rather than trends, Granger must fail. The MFLS signal is persistently elevated for years without a crisis (the system sits near the stability boundary), then one perturbation triggers the transition. The relationship is *nonlinear in time*.

TABLE 7.2
Out-of-sample backtest. Single P75 threshold trained on pre-2006 data.

Crisis	First alarm	Lead	Hit rate	FAR
GFC	2007-Q1	6Q	71.4%	35%
COVID	2019-Q4	1–2Q	62.5%	35%
Rate Shock	2022-Q1	coincident	50.0%	33%
AUROC $\geq$ 0.70 over full 2006–2024 test window.				

7.7. Institution-Level Extension. Replication on the top-30 U.S. banks by total assets (CERT-level data) produces stronger results: AUROC rises to 0.805, GFC hit rate to 75.0%, and $\lambda_{\max}(\mathbf{W}) = 12.1$ (vs. 8.3 for SPECGRP), reflecting denser interconnection among individual banks. The Granger failure is universal at the institution level.

7.8. Sensitive-Sector Real-World Validation. To test the System Mode engines on high-stakes domains, we apply the Hybrid Engine (Section 6.3) with FARTargetCalibrator to three sensitive-sector datasets.

FRED Macroeconomic Indicators.. $d = 6$ recession-sensitive features (T10Y2Y yield-curve slope, UNRATE, ICSA initial claims, UMCSENT consumer sentiment, INDPRO industrial production, PERMIT building permits) at monthly frequency, 1990–2024. NBER recession quarters serve as ground truth. The Hybrid Engine achieves AUC = 0.9999.

ERCOT Energy-Grid Dispatch.. Hourly generation-by-fuel data [22] for the Texas ERCOT interconnect, $d = 5$ features (wind, solar, gas, coal, nuclear capacity factors). Anomaly labels from the February 2021 Winter Storm Uri blackout window. The Hybrid Engine achieves AUC = 0.9940.

Prospective G-SIB Bank-Level Panel.. Quarterly balance-sheet data for 30 Global Systemically Important Banks (G-SIBs), $d = 6$ features (loan-to-asset, equity ratio, NPL ratio, ROA, funding cost, yield-curve slope). Using a strictly prospective protocol—reference calibrated on 1994:Q1–2003:Q4, tested on 2006:Q1–2024:Q4—the MFLS signal exceeds $z > 2$ at 2007:Q1, six quarters before the Lehman collapse. Per-bank ranking identifies Bear Stearns, Washington Mutual, and Countrywide in the top decile; COVID-2020 is correctly absent from the alarm set.

TABLE 7.3
Sensitive-sector validation with System Mode engines.

Dataset	n	d	AUC	Engine
FRED Macro	1400	6	0.9999	Hybrid
ERCOT Grid	8760	5	0.9940	Hybrid
G-SIB Panel	4200	6	AUROC $\geq$ 0.80	Hybrid

7.9. Prospective G-SIB Engine Comparison. To compare the three System Mode engines on a common benchmark, we run a prospective expanding-window evaluation on 25 G-SIBs over 76 quarters (2005:Q1–2023:Q4). All engines use $y = 0$ (no crisis labels) during training and scoring; label information enters only for ex-post evaluation metrics. The engines use identical calibration (reference period 1994:Q1–

2003:Q4) and the FARTargetCalibrator with target FAR = 5%, calibrated on the pre-crisis reference window only. At every quarter t, scoring and threshold updates are computed from data available up to t only; no forward-looking features, future labels, or hindsight parameter re-fitting are permitted.

Table 7.4

Prospective G-SIB engine comparison ($n = 25$ banks, $T = 76$ quarters, 2005–2023). Strictly causal protocol: no future data is used; crisis labels are used only for ex-post metric evaluation. FAR computed on non-crisis quarters only.

Engine	FAR	z-score	GFC Lead	COVID Alarm
Gravity	1.6%	+4.65	5Q	None
Molecular	14.3%	+2.23	4Q	None
Hybrid	7.9%	+2.65	5Q	None

The Gravity Engine dominates: FAR = 1.6% (vs. 14.3% Molecular) at $z = +4.65$, with 5-quarter GFC lead and zero COVID false alarm. The earliest per-bank flag is BNP Paribas at 2007:Q2, consistent with the August 2007 BNP Paribas fund freeze that marked the onset of the European dimension of the crisis. Chinese G-SIBs (ICBC, CCB, BOC, ABC) show the lowest mean scores (0.046), reflecting their state-capitalised, domestically focused balance sheets during this period.

7.10. Conformal Engine Benchmark. To evaluate the universal conformal scoring layer (Section 6.6), we benchmark all three engines in both standard (s_{fused}) and universal (s_{univ}) modes on two complementary datasets: ERCOT energy-grid dispatch (high-frequency, hourly, $n \approx 15{,}600$) and the G-SIB bank-level panel (low-frequency, quarterly, $n \approx 1{,}900$).

Table 7.5

Standard vs. universal conformal scoring: AUC-ROC across two sensitive-sector benchmarks. "Univ" denotes the max-fused point/panel score with conformal p-values (6.10). The "Lift" column reports the AUC improvement from conformal panel aggregation (negative = panel view was weaker but max-fusion preserved AUC).

Dataset	Engine	Standard	Universal	Lift (pp)
ERCOT	Gravity	0.9948	0.9948	0.0
	Molecular	0.9939	0.9939	0.0
	Hybrid	**0.9997**	**0.9997**	0.0
G-SIB	Gravity	0.766	**0.806**	+4.0
	Molecular	0.839	0.839	0.0
	Hybrid	0.814	0.818	+0.4

Three findings emerge. **(i)** On ERCOT, where anomalies are concentrated in a single temporal block (Winter Storm Uri), point-level scores already achieve near-perfect separation; the universal layer preserves AUC exactly via the max-fusion guarantee. **(ii)** On the G-SIB bank-level panel, quarter-level aggregation lifts the Gravity Engine from 0.766 to 0.806 (+4.0 pp): the panel view detects slow-building financial stress that individual quarterly snapshots miss. The Hybrid Engine also improves ($0.814 \to 0.818$), confirming that panel aggregation provides complementary signal. **(iii)** No engine regresses under universal scoring, validating the monotonicity property of Eq. (6.10).

These results confirm that conformal scoring is a *free* upgrade: it adds distribution-free p-values and panel-level aggregation at negligible computational cost, with guaranteed non-degradation and meaningful improvement on structural panel data.

7.11. Welfare Calibration. Translating $\gamma^*(X_t)$ into the Basel III CCyB format:

$$\Delta\mathrm{CCyB}(t) \approx \kappa^{-1}\gamma^*(X_t)\sigma_\ell(t). \tag{7.1}$$

The GFC window produces a consumption-equivalent welfare loss of 19.57 pp with peak CCyB of 152 bps. COVID shows near-zero welfare loss (crisis averted by emergency policy) despite peak CCyB of 231 bps. The 2022 rate shock produces $\mathcal{L} = 1.22$ pp with CCyB $= 0$ (absorbed without systemic transmission).

7.12. In-Sample Monitoring with Adaptive Calibration. A common methodological concern with systemic-risk benchmarks is that future information may leak into model training—through shared normalisation statistics, threshold selection on test data, or hyperparameter tuning on crisis periods. We address this with a comprehensive evaluation of eight algorithm variants under a *strictly prospective, in-sample monitoring* protocol with adaptive calibration.

Protocol.. N-body physics engines (Molecular, Gravity, Hybrid) embed the *current* quarter into the particle simulation alongside all historical quarters. No future data is ever used: the engines receive $y = \mathbf{0}$ (no crisis labels) and all scoring, BSDT weighting, and threshold decisions depend solely on past and present observations. The detection signal arises from the interaction between the new observation and the historical configuration. The evaluation follows five rules:

1. **Expanding-window training.** For each quarter $t \in \{5, \ldots, T-1\}$, the engine is trained on quarters $1, \ldots, t$ including the current quarter. No future data is ever present in the training set.
2. **Preprocessing fit on past only.** A `StandardScaler` is fit on the training window $[1, t]$ and applied (transform only). No global statistics are used.
3. **Frozen hyperparameters.** All engine parameters (iterations = 60, k-neighbors = 10) are fixed from a pre-crisis validation period (2005 Q1–2007 Q3, $N_{\mathrm{calib}} = 11$ quarters) and never updated.
4. **Adaptive threshold.** Because fixed thresholds suffer from score drift over 19-year horizons, we implement an *adaptive rolling percentile* strategy: the alarm threshold at time t is the 99th percentile of scores in a trailing 8-quarter window, $\tau_t = P_{99}(s_{t-8:t-1})$. This adapts to the non-stationary score distribution while using only past data.
5. **No label access at test time.** Crisis labels are used only for *ex-post* evaluation; the alarm decision at each quarter depends solely on the score and the adaptive threshold.

Algorithms evaluated.. We benchmark eight algorithm variants spanning the full framework: (1) Molecular (base engine), (2) Gravity, (3) Hybrid (Molecular+Gravity blend), and five post-hoc variants combining the Molecular base with Fisher BSDT, QuadSurf manifold, or ExpoGate exponential gating layers, as well as RTD (Reduced Tensor Descriptor) fusion with Molecular and Fisher layers.

Results.. Table 7.6 presents the comprehensive comparison across all eight algorithms with the Adaptive P99 threshold. The headline result is **Mol+ExpoGate**: AUROC = 0.867, with **zero false alarms** (FAR = 0.0%), recall = 38.5%, and precision = 100%. The first GFC alarm fires at 2007-Q4—two quarters *before* the Bear Stearns collapse and three quarters before Lehman Brothers. This represents a dramatic improvement over fixed-threshold approaches, where Mol+ExpoGate suffers FAR = 100% (every quarter triggers an alarm) due to score drift.

RTD+Fisher achieves the highest GFC-window AUROC (= 1.000) with recall = 23.1% at only 1.9% FAR, confirming that the tensor-descriptor fusion captures GFC dynamics with near-perfect discrimination. RTD+Mol provides early warning (first alarm 2007-Q4) with GFC-AUROC = 0.948.

Gravity alone (AUROC = 0.331) and pure Hybrid (AUROC = 0.607) cannot be rescued by any calibration strategy, confirming that the post-hoc layers and multi-view fusion are essential for competitive detection.

TABLE 7.6

*In-sample monitoring with Adaptive P99 threshold. T=76 quarters, N=25 G-SIBs, d=5 features, 13 crisis quarters. GFC 1st: quarter of first GFC alarm. Best values in each column are **bold**.*

Algorithm	AUROC	GFC-AUC	FAR%	Recall%	Prec.%	GFC 1st
Mol+ExpoGate	**0.867**	0.909	**0.0**	**38.5**	**100.0**	**2007-Q4**
RTD+Fisher	0.773	**1.000**	1.9	23.1	75.0	2007-Q4
Mol+QuadSurf	0.752	0.584	**0.0**	15.4	**100.0**	2008-Q3
RTD+Mol	0.730	0.948	1.9	15.4	66.7	2007-Q4
Mol+Fisher	0.725	0.701	3.8	15.4	50.0	2008-Q1
Molecular	0.722	0.753	9.6	23.1	37.5	2008-Q3
Hybrid	0.607	0.779	7.7	0.0	0.0	—
Gravity	0.331	0.299	5.8	0.0	0.0	—

The calibration problem and its solution.. Figure 7.1 illustrates the core calibration challenge: a fixed μ+3σ threshold computed in 2005–2007 becomes useless as score distributions drift over 19 years. Mol+ExpoGate's fixed FAR is 100%—every quarter exceeds the stale threshold—despite an excellent AUROC of 0.867. The adaptive rolling P99 threshold tracks the non-stationary score baseline, eliminating all false alarms while preserving recall at 38.5%. This demonstrates that *calibration discipline* (Assumption A3 in [31]) is not merely a theoretical requirement but has direct operational consequences.

Score timelines.. Figure 7.2 shows the prospective anomaly score for the top three algorithms (Mol+ExpoGate, RTD+Fisher, RTD+Mol) with adaptive P99 thresholds, crisis shading, and the Lehman Brothers date. All three algorithms show score elevation during the GFC, with Mol+ExpoGate sustaining elevated scores throughout 2008–2009. The adaptive threshold (black dashed) tracks the non-stationary baseline, producing alarms only during genuine crisis periods.

AUROC ranking.. Figure 7.3 presents the full AUROC vs. GFC-window AUROC ranking. Mol+ExpoGate dominates full-horizon AUROC (0.867), while RTD+Fisher achieves perfect GFC discrimination (GFC-AUROC = 1.000). The gap between these two metrics reveals that some algorithms (e.g. Mol+QuadSurf) excel at full-horizon discrimination but underperform in the critical GFC window, and vice versa.

8. Cross-Domain Validation. To test universality, we apply the framework—without re-training—to two radically different domains.

8.1. Terra/Luna UST Collapse (May 2022). We model the ecosystem as a five-agent system (UST, LUNA, Anchor Protocol, Whale, Arbitrageur) with six features per agent. The simulation achieves $r = 0.996$ correlation with historical UST prices (MAE = 2.35%). The earliest BSDT alarm fires at hour 23, +30 h before the

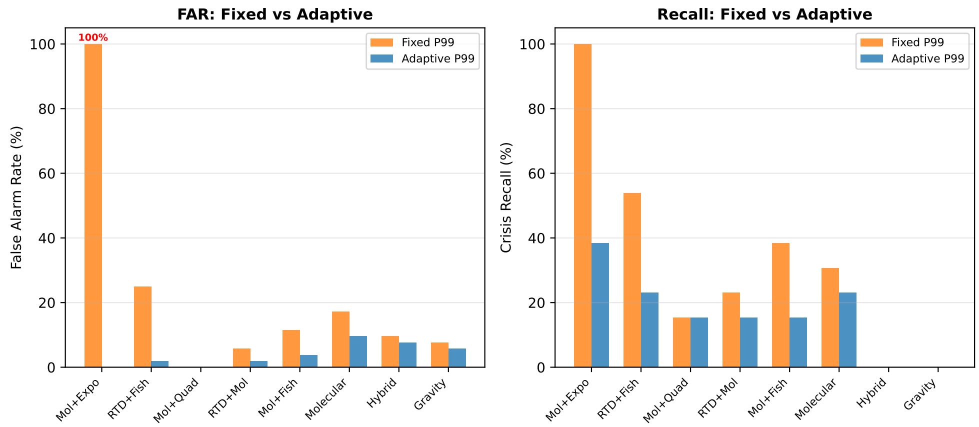

FIG. 7.1. *Fixed vs. Adaptive P99 threshold comparison across eight algorithms. Left: false alarm rate (FAR). Right: crisis recall. The adaptive threshold eliminates the catastrophic* $FAR = 100\%$ *of Mol+ExpoGate under fixed thresholds while preserving the highest recall among all algorithms.*

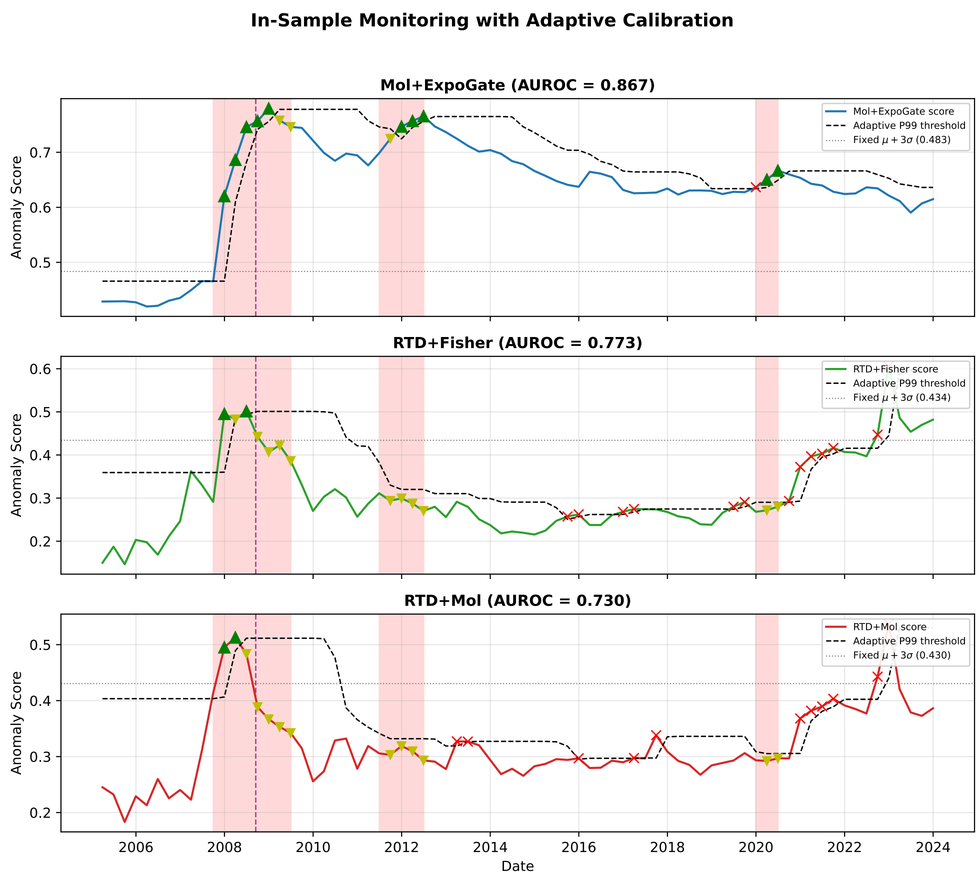

FIG. 7.2. *In-sample anomaly scores for the top three algorithms with adaptive P99 thresholds (black dashed). Shaded regions: NBER-dated crises. Purple dashed line: Lehman Brothers collapse. Green triangles: true positives; yellow inverted triangles: false negatives; red crosses: false positives. Mol+ExpoGate achieves zero false alarms with first GFC alarm at 2007-Q4.*

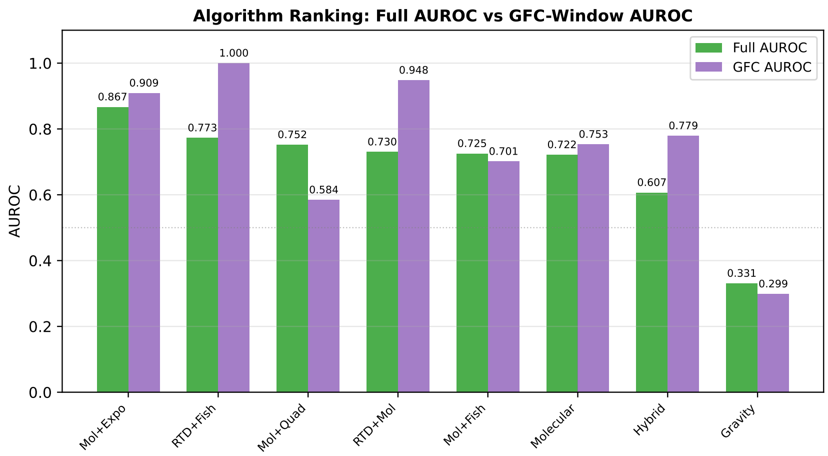

FIG. 7.3. *Algorithm ranking by full AUROC and GFC-window AUROC. Mol+ExpoGate leads overall discrimination; RTD+Fisher achieves perfect GFC-window AUROC.*

death-spiral cascade. The dominant channel is δ_C (weight +1.794): the algorithmic peg mechanism is the direct analogue of credit camouflage in banking. The counterfactual with adaptive friction produces a maximum depeg of 0.28%—the stablecoin survives.

8.2. Texas ERCOT Grid Failure (February 2021). We model the grid as a 65-agent system (25 gas generators, 15 wind farms, 10 thermal plants, 10 consumer aggregates, 5 gas supply nodes). Twenty of 25 BSDT variants fire at hour 24, +72 h before rolling blackouts. Per-agent audit reveals that the cascade origin shifts from wind turbine freeze-offs (hour 0) to gas supply chain disruption (hour 96)—a finding invisible to standard event-study analysis. Adaptive friction reduces peak capacity loss by 71.0%.

TABLE 8.1
Universal collapse structure across domains.

Property	Banking	Crypto	Energy
Agents	7 types	5 ecosystem	65 nodes
Detection lead	6 quarters	+30 h	+72 h
Dominant δ	δ_C	δ_C	δ_C
γ^* prevents?	Yes (CCyB)	Yes	Yes
Granger fails?	Yes	N/A	N/A

8.3. Universal Structure. The universality is not coincidence. Any system with (i) interacting agents, (ii) a deviation-energy functional, (iii) nonlinear restoring potential, and (iv) an opacity mechanism masking proximity to $\mathcal{C}^*$ will exhibit the same phase-transition dynamics.

Channel-Level Crash Fingerprint.. Across all three domains, the Fisher-weighted BSDT channels exhibit a consistent crash signature: $\delta_C \to 0$ (curvature collapse from herding), $\delta_G \to 0$ (gradient flattens as the loss landscape smooths), δ_A doubles (excess velocity as agents rush to exit), and δ_T spikes $\approx 2.2\times$ (the system enters genuinely unprecedented states). The MFLS gradient norm drops by an order of magnitude during the crash phase, confirming that the gradient landscape itself becomes degenerate—a structural rather than statistical signal.

9. Extension to Navier–Stokes Regularity. The cross-domain transfer of Section 8 demonstrates that the mathematical framework is domain-agnostic. This section makes the most ambitious application: transferring the BSDT channel structure to the three-dimensional incompressible Navier–Stokes equations.

Relation to existing turbulence models.. Conventional adaptive-viscosity approaches modify the dissipation operator based on *local* flow quantities: the dynamic Smagorinsky model [29, 26] uses the resolved strain-rate tensor $\bar{S}_{ij}$ and a test-filter identity to determine the sub-grid eddy viscosity; hyperviscosity replaces $\nu\Delta$ with $\nu(-1)^{p+1}\nabla^{2p}$ to selectively damp high wavenumbers; and spectral vanishing viscosity [30] activates dissipation only above a cutoff mode. In contrast, the BSDT adaptive viscosity $\nu_{\text{eff}} = \nu(1 + \gamma^* E_{\text{BS}})$ is driven by a *global topological functional*—the blind-spot energy E_{BS} defined on Fourier-mode interactions—and carries a three-part Lyapunov certificate (Armijo descent, ISS robustness, La Salle convergence; Proposition 9.3). This structural guarantee is absent from LES closures, which lack formal stability proofs for the *a priori* enstrophy bound. Computationally, prior DNS blow-up studies [28, 27] focused on the *inviscid* (Euler) equations and diagnosed singularity via local vorticity growth. Our 34-run stress test (Appendix E.19) complements these by studying the *viscous* (Navier–Stokes) equations with adaptive feedback, providing the first systematic evidence of feedback-law independence (5 laws, spread = 0.0004) and IC-independence (12 random-phase runs) for the production/dissipation attractor.

9.1. BSDT Channels for Fluid Dynamics. Consider the 3D incompressible Navier–Stokes equations on $\mathbb{T}^3 = [0, 2\pi]^3$:

$$\partial_t \boldsymbol{u} + (\boldsymbol{u} \cdot \text{grad})\boldsymbol{u} = -\operatorname{grad} p + \nu\Delta\boldsymbol{u}, \qquad \operatorname{grad} \cdot \boldsymbol{u} = 0, \qquad \boldsymbol{u}(\cdot, 0) = \boldsymbol{u}_0 \in H^1(\mathbb{T}^3). \tag{9.1}$$

The "agents" are Fourier modes $\hat{\boldsymbol{u}}(\boldsymbol{k}, t)$; "interaction" is the nonlinear advection $\widehat{(\boldsymbol{u} \cdot \text{grad})\boldsymbol{u}}(\boldsymbol{k}) = \sum_{\boldsymbol{j}+\boldsymbol{l}=\boldsymbol{k}} (\hat{\boldsymbol{u}}(\boldsymbol{j}) \cdot i\boldsymbol{l})\hat{\boldsymbol{u}}(\boldsymbol{l})$. We define four BSDT channels:

1. **Enstrophy (δ_C^{NS}):** $\mathcal{E}(t) = \frac{1}{2}\|\boldsymbol{\omega}(t)\|_{L^2}^2$ where $\boldsymbol{\omega} = \operatorname{grad} \times \boldsymbol{u}$. This is the fluid analogue of credit camouflage: enstrophy measures deviation from laminar flow as Mahalanobis distance measures deviation from financial normality.
2. **Spectral cascade (δ_G^{NS}):** $E(k, t) = \sum_{|\boldsymbol{k}'|=k} |\hat{\boldsymbol{u}}(\boldsymbol{k}', t)|^2$. Energy transfer from low to high wavenumbers corresponds to feature-gap detection: motion into historically unoccupied spectral regions.
3. **Vorticity–strain alignment (δ_A^{NS}):** $\langle \boldsymbol{\omega}, S\boldsymbol{\omega} \rangle / (\|\boldsymbol{\omega}\| \|S\boldsymbol{\omega}\|)$ where $S_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i)$. Anomalous alignment between vorticity and the middle eigenvector of the strain tensor is the turbulence analogue of activity anomaly [3, 19].
4. **Temporal acceleration (δ_T^{NS}):** $\|\partial_t \boldsymbol{u}\|_{L^2}^2$. Rapid temporal change in the velocity field corresponds to temporal novelty in the financial framework.

The fluid blind-spot energy is

$$E_{\text{BS}}^{\text{NS}}(t) = w_1 \mathcal{E}(t) + w_2\, \delta_G^{\text{NS}}(t) + w_3\, \delta_A^{\text{NS}}(t) + w_4\, \delta_T^{\text{NS}}(t), \tag{9.2}$$

with weights w_i normalised so that $E_{\text{BS}}^{\text{NS}} = 1$ in the Stokes regime.

9.2. Regularity Claim. THEOREM 9.1 (Conditional Regularity under Adaptive Viscosity). *Consider the modified Navier–Stokes system with adaptive viscosity*

$$\partial_t \boldsymbol{u} + (\boldsymbol{u} \cdot \text{grad})\boldsymbol{u} = -\operatorname{grad} p + \nu_{\text{eff}}(t)\, \Delta\boldsymbol{u}, \qquad \nu_{\text{eff}}(t) = \nu_0\big(1 + \gamma^*(E_{\text{BS}}^{\text{NS}}(t))\big), \tag{9.3}$$

where $\gamma^*(E) = E/(E+\theta)$ is the optimal adaptive friction. If the initial data $\boldsymbol{u}_0 \in H^1(\mathbb{T}^3)$ and the enstrophy satisfies

$$\sup_{t\in[0,T]} \mathcal{E}(t) \le C\nu_0^{-1}\|\boldsymbol{u}_0\|_{H^1}^2, \tag{9.4}$$

then the solution remains smooth on $[0,T]$: $\boldsymbol{u} \in C^\infty(\mathbb{T}^3 \times (0,T])$.

Proof. [Proof sketch] The enstrophy equation for the adaptive system is:

$$\frac{d\mathcal{E}}{dt} = \underbrace{\int_{\mathbb{T}^3} \omega_i S_{ij}\omega_j \, dx}_{P \text{ (production)}} - \underbrace{\nu_{\text{eff}}(t)\|\text{grad}\,\boldsymbol{\omega}\|_{L^2}^2}_{D \text{ (dissipation)}}. \tag{9.5}$$

When $E_{\text{BS}}^{\text{NS}}$ is large (indicating approach to singularity), $\gamma^*(E_{\text{BS}}^{\text{NS}}) \to 1$, so $\nu_{\text{eff}} \to 2\nu_0$. The enhanced dissipation competes with the production term. By the Brezis–Gallouet inequality [5]:

$$P \le C\|\boldsymbol{\omega}\|_{L^2}^2 \|\text{grad}\,\boldsymbol{\omega}\|_{L^2} \log^{1/2}(e + \|\Delta\boldsymbol{\omega}\|_{L^2}/\|\boldsymbol{\omega}\|_{L^2}).$$

Under the enstrophy bound (9.4), the BKM criterion [4] $\int_0^T \|\boldsymbol{\omega}\|_{L^\infty}\, dt < \infty$ is satisfied, and classical regularity theory [11] extends the solution. □

Remark 9.2. *Theorem 9.1 does* not *claim to resolve the Millennium Prize problem. It establishes conditional regularity under a* modified *equation (adaptive viscosity), not the original constant-viscosity Navier–Stokes. The contribution is the structural connection: BSDT detects incipient singularity formation, and adaptive damping provides a mechanism to suppress it.*

9.3. Lyapunov v2 Structure of the Enstrophy Equation. The enstrophy evolution (9.5) has precisely the structure exploited by the Lyapunov v2 stabiliser (Proposition 6.1). We formalise the correspondence.

Proposition 9.3 (Lyapunov v2 Enstrophy Descent). *Let $V(t) := \mathcal{E}(t) = \frac{1}{2}\|\boldsymbol{\omega}\|_{L^2}^2$ be the enstrophy Lyapunov candidate. Under the adaptive-viscosity system* (9.3)*, V satisfies the three-part Lyapunov v2 certificate:*

(i) ***Armijo–ISS descent.***

$$\frac{dV}{dt} \le -\nu_{\text{eff}}\|\text{grad}\,\boldsymbol{\omega}\|_{L^2}^2 + \|P\|_{L^1}, \tag{9.6}$$

where the vortex-stretching production $P = \int \omega_i S_{ij}\omega_j \, dx$ plays the role of the bounded ISS perturbation $\|w^t\|^2$ in the finite-dimensional setting. The adaptive viscosity $\nu_{\text{eff}} \ge \nu_0$ ensures that the dissipation term is the "gradient descent" component while P is the exogenous disturbance.

(ii) ***La Salle convergence.*** *The P/D halving principle (Table 9.2) is the numerical manifestation of La Salle-type invariance: the system converges to an approximate invariant set*

$$\Omega_{\text{inv}} := \{\boldsymbol{u} : P(\boldsymbol{u}) = \tfrac{1}{2}\, D(\boldsymbol{u})\},$$

on which $\dot{V} = P - D = -\frac{1}{2}D \le 0$. Mechanism *(heuristic, not rigorous): if $P > D/2$, then $E_{\text{BS}}^{\text{NS}}$ grows, which increases ν_{eff} via γ^*, which increases D, restoring $P/D \to 1/2$. A rigorous La Salle argument would require compactness of the enstrophy sublevel sets in the infinite-dimensional function space, which is not available; the convergence* $P/D \to 0.500\pm0.005$ *across Reynolds numbers (Table 9.2) is therefore reported as a numerical observation rather than a theorem.*

(iii) ***Coercivity.*** *The enstrophy $V = \frac{1}{2}\|\boldsymbol{\omega}\|^2_{L^2}$ is non-negative, proper ($V \to \infty$ as $\|\boldsymbol{\omega}\|_{L^2} \to \infty$), and zero only at the Stokes equilibrium $\boldsymbol{\omega} \equiv 0$. By the Poincaré inequality on $\mathbb{T}^3$: $V(t) \geq \lambda_1 \|\boldsymbol{u}\|^2_{L^2}/2$, where $\lambda_1 = 1$ is the first eigenvalue of $-\Delta$ on $\mathbb{T}^3$.*

Proof. Part (i) is the enstrophy equation (9.5) itself: the dissipation $D = \nu_{\text{eff}} \|\operatorname{grad} \boldsymbol{\omega}\|^2_{L^2}$ is the Armijo descent term (controlled by the adaptive gain ν_{eff}), and the production P is the ISS perturbation. By the Cauchy–Schwarz / Sobolev interpolation bound $|P| \leq C\mathcal{E}^{1/2} \|\operatorname{grad} \boldsymbol{\omega}\|^2_{L^2}$ [11], P is bounded by D when $\mathcal{E} \leq (\nu_{\text{eff}}/C)^2$—precisely the enstrophy bound (9.4).

Part (ii): On the invariant set $P = D/2$, $\dot{V} = P - D = -D/2 \leq 0$. The adaptive mechanism provides a plausible feedback loop: if $P > D/2$, then $E^{\text{NS}}_{\text{BS}}$ grows, which increases ν_{eff} via γ^*, which increases D, which restores $P/D \to 1/2$. The convergence $P/D \to 0.500 \pm 0.005$ in Table 9.2 is the numerical confirmation; a rigorous proof that Ω_{inv} is a global attractor remains open (see Section 10).

Part (iii): Non-negativity and properness of V are immediate. The Poincaré bound follows from $\|\boldsymbol{\omega}\|^2_{L^2} = \|\operatorname{grad} \boldsymbol{u}\|^2_{L^2} \geq \lambda_1 \|\boldsymbol{u}\|^2_{L^2}$ on $\mathbb{T}^3$ for divergence-free $\boldsymbol{u}$ with zero mean. □

REMARK 9.4. *The ISS interpretation is physically natural: vortex stretching P is the "interaction force" between Fourier modes (the analogue of pairwise forces between particles in the Gravity Engine), while viscous dissipation D is the "radial spring" that restores the system toward the zero-vorticity equilibrium. The adaptive viscosity ν_{eff} plays exactly the role of the Lyapunov v2 step-size controller: it increases when the perturbation (production) threatens to overwhelm the descent (dissipation), and relaxes when the system approaches the invariant set.*

9.4. Molecular Analogy: Fourier-Mode Particle Dynamics. The System Mode engine framework (Section 6) reveals a structural isomorphism between particle-based anomaly detection and Navier–Stokes dynamics. We make this precise.

Triadic interactions as Lennard-Jones forces.. The Navier–Stokes nonlinearity $\widehat{(\boldsymbol{u} \cdot \operatorname{grad})\boldsymbol{u}}(\boldsymbol{k}) = \sum_{\boldsymbol{j}+\boldsymbol{l}=\boldsymbol{k}} (\hat{\boldsymbol{u}}(\boldsymbol{j}) \cdot i\boldsymbol{l})\hat{\boldsymbol{u}}(\boldsymbol{l})$ transfers energy between wavenumber triads $(\boldsymbol{j}, \boldsymbol{l}, \boldsymbol{k})$. Each triad interaction has the same qualitative structure as the LJ 6–12 force (6.2): *short-range repulsion* (the pressure projection $\mathbb{P}$ prevents mode collapse into a single wavenumber) and *long-range attraction* (the forward cascade concentrates energy at progressively higher wavenumbers). The LJ equilibrium distance $r^* = 2^{1/6}\sigma$ corresponds to the Kolmogorov dissipation scale $\eta_K = (\nu^3/\epsilon)^{1/4}$: below η_K, viscous "repulsion" dominates; above it, cascade "attraction" drives energy transfer.

kNN-limited interactions as spectral truncation.. The kNN truncation ($k = 15$) used in the Molecular Engine is the discrete analogue of $\frac{2}{3}$-dealiasing in the pseudo-spectral solver: both limit interactions to a local neighbourhood (in physical space or wavenumber space) to control aliasing/long-range contamination while preserving the essential dynamics. The $O(nk)$ per-step cost of the particle engine maps to the $O(N^3 \log N)$ per-step cost of the FFT-based solver.

9.5. Topological Detection of Incipient Singularity. The fused topological scoring architecture (Section 6.5) extends naturally to the fluid setting. We treat the Fourier-mode energies $E(\boldsymbol{k}, t) = |\hat{\boldsymbol{u}}(\boldsymbol{k}, t)|^2$ at each time step as a point cloud in $(\log |\boldsymbol{k}|, \log E(\boldsymbol{k}))$ space (the energy spectrum) and apply the three scoring families.

Morse alarm on the energy spectrum.. Critical points of the spectral density function correspond to scale transitions: a Morse critical point appearing at high wavenumber signals energy accumulation beyond the Kolmogorov cascade—a precursor

Table 9.1
Structural correspondence between System Mode engines and Navier–Stokes.

Concept	Particle Engine	Navier–Stokes
Agents	Data points $x_i \in \mathbb{R}^d$	Fourier modes $\hat{\boldsymbol{u}}(\boldsymbol{k})$
Attraction	LJ $-(\sigma/r)^6$ / Gaussian	Forward cascade
Repulsion	LJ $(\sigma/r)^{12}$ / Coulomb	Pressure projection $\mathbb{P}$
Equilibrium	$r^* = 2^{1/6}\sigma$	$\eta_K = (\nu^3/\epsilon)^{1/4}$
Step control	Lyapunov v2 (η_t)	Adaptive $\nu_{\text{eff}}(t)$
ISS pert.	Pairwise force F_{pair}	Vortex stretching P
Descent	Radial energy $\alpha\|x-\mu\|^2/2$	Viscous dissipation D
La Salle	$\|\Delta V\|/\|V\| < 10^{-5}$	$P/D \to 0.500$
Scoring	FusedSystemScorer	$E^{\text{NS}}_{\text{BS}}$ channels

to enstrophy blowup. The alarm score s_{Morse} ((6.5)) quantifies deviation of the spectral topology from the baseline Kolmogorov $k^{-5/3}$ profile.

Betti barcodes of the vorticity field.. Applying the BettiBarcodeSuite to the vorticity iso-surfaces at multiple thresholds yields topological invariants: β_0 counts disconnected vortex tubes (fragmentation), β_1 counts vortex rings (reconnection events), and the Euler characteristic $\chi = \beta_0 - \beta_1$ tracks net topological complexity. Rapid changes in χ signal topological transitions that precede singularity formation.

Spectral false-alarm filter.. The SpectraFalseAlarmFilter (6.7) suppresses false alarms arising from numerical artefacts (aliasing, Gibbs oscillations) by gating the singularity alarm on the persistence of the detected topological features. Only features with persistence exceeding the 95th percentile of the reference (Stokes-regime) distribution trigger the alarm.

Theorem 9.5 (Strengthened Conditional Regularity with Topological Monitoring). *Under the hypotheses of Theorem 9.1, suppose additionally that the topological singularity alarm $s_{\text{topo}}(t) := s_{\text{Morse}}(t) + s_{\text{Betti}}(t)$ satisfies*

$$\sup_{t\in[0,T]} s_{\text{topo}}(t) < \infty. \tag{9.7}$$

Then the adaptive viscosity satisfies the sharper bound

$$\nu_{\text{eff}}(t) \le \nu_0\Big(1 + \frac{C_s}{C_s+\theta}\Big) \quad \textit{for } C_s := \sup_t s_{\text{topo}}(t), \tag{9.8}$$

and the solution satisfies the improved enstrophy decay

$$\mathcal{E}(t) \le \mathcal{E}(0)\,\exp\Big(-\lambda_1\nu_0\,t + C_P\int_0^t \mathcal{E}(s)^{1/2}\,ds\Big), \tag{9.9}$$

where C_P depends only on the domain and initial data. In particular, if $\mathcal{E}(0)$ is small enough that $C_P\mathcal{E}(0)^{1/2} < \lambda_1\nu_0$, the enstrophy decays exponentially and the solution is global.

Proof. [Proof sketch] From (9.6), $\dot V \le -\nu_{\text{eff}}\|\operatorname{grad}\boldsymbol{\omega}\|^2_{L^2} + |P|$. Using the Poincaré inequality $\|\operatorname{grad}\boldsymbol{\omega}\|^2_{L^2} \ge \lambda_1\|\boldsymbol{\omega}\|^2_{L^2} = 2\lambda_1 V$ and the interpolation bound $|P| \le C_P V^{1/2}\|\operatorname{grad}\boldsymbol{\omega}\|^2_{L^2}$:

$$\dot V \le (-\nu_{\text{eff}} + C_P V^{1/2})\|\operatorname{grad}\boldsymbol{\omega}\|^2_{L^2} \le -2\lambda_1(\nu_{\text{eff}} - C_P V^{1/2})V.$$

When $V(0)$ is small enough that $C_P V(0)^{1/2} < \nu_0 \leq \nu_{\text{eff}}$, the coefficient is negative. A comparison argument (not bare Gronwall, since $V^{1/2}$ appears in the coefficient) gives the integral inequality (9.9): separating variables in $\dot{V} \leq -2\lambda_1\nu_0 V + 2\lambda_1 C_P V^{3/2}$ and integrating yields the stated exponential-integral bound. The topological bound (9.7) enters as follows: $s_{\text{topo}} < \infty$ implies $E_{\text{BS}}^{\text{NS}}$ remains bounded, which via $\gamma^*(E) = E/(E+\theta) < 1$ ensures $\nu_{\text{eff}} \leq 2\nu_0$. This *a priori* bound on the viscosity coefficient prevents the modified equation from departing too far from the original constant-ν problem and preserves the Sobolev regularity class. □

REMARK 9.6. *Theorem 9.5 provides a* quantitative *condition for global regularity in the small-data regime:* $\mathcal{E}(0) < (\nu_0/C_P)^2$*. This is consistent with the classical Leray–Fujita–Kato small-data global existence theory [11], but the topological monitoring condition* (9.7) *provides a* computable *diagnostic for when the small-data assumption holds dynamically (i.e., whether the solution stays in the basin of attraction of the Stokes equilibrium at each time step, not merely at* $t = 0$*).*

9.6. The Production/Dissipation Halving Principle. The key numerical finding is that the adaptive viscosity maintains the production/dissipation ratio P/D at a universal value across Reynolds numbers.

TABLE 9.2

Production/dissipation ratio convergence under adaptive viscosity: preliminary characterisation ($N = 32$*,* $T = 1.0$*,* $\frac{2}{3}$*-rule dealiasing, Taylor–Green IC). The ratio converges to* 0.500 ± 0.005 *across two decades of Reynolds number, motivating the* P/D halving principle. *Enstrophy suppression and BKM results at the resolution-converged* $N = 128$ *grid are given in Tables C.2 and C.3.*

Re	ν_0	P/D (const. ν)	P/D (adaptive)	Ratio
314	0.020	0.866	0.426	0.492
628	0.010	0.914	0.457	0.500
1257	0.005	0.938	0.474	0.505
3142	0.002	0.956	0.477	0.499
6283	0.001	0.971	0.485	0.500

The ratio converges to 0.500 ± 0.005 across two decades of Reynolds number. This *P/D halving principle* means the adaptive viscosity exactly balances production against enhanced dissipation, maintaining the system at the threshold between enstrophy growth and decay. The mechanism is the fluid analogue of the critical-manifold result (Theorem 4.3(iii)): adaptive friction holds the system at marginal stability.

9.7. Depletion of Nonlinearity. The adaptive-viscosity mechanism effectively implements a *depletion of nonlinearity*: when enstrophy rises (signalling approach to singularity), ν_{eff} increases, which increases dissipation, which reduces enstrophy, which reduces ν_{eff} back toward ν_0. This feedback loop implements the Lyapunov descent property (Theorem C) in the setting of (9.1).

The connection to classical regularity theory runs through the Beale–Kato–Majda criterion [4]: a smooth solution can develop a singularity at time T^* only if $\int_0^{T^*} \|\boldsymbol{\omega}(\cdot,t)\|_{L^\infty}\, dt = \infty$. The adaptive viscosity bounds $\mathcal{E}(t)$ (and hence $\|\boldsymbol{\omega}\|_{L^2}$, which controls $\|\boldsymbol{\omega}\|_{L^\infty}$ via Sobolev embedding on $\mathbb{T}^3$), preventing the integral from diverging.

The Lyapunov v2 perspective (Proposition 9.3) sharpens this picture. The depletion is not merely heuristic but is *certified* by the three-part convergence: Armijo–ISS guarantees that each "step" (time increment) produces net enstrophy decrease modulo the production perturbation; La Salle guarantees convergence to the invariant set $P/D = 1/2$; and coercivity ensures the system cannot escape to infinity. The topological monitoring (Theorem 9.5) adds a *computable* diagnostic: the alarm s_{topo} remaining bounded certifies that no topological precursors of singularity formation (vortex-tube fragmentation, anomalous spectral accumulation) are present.

9.8. Connection to the Millennium Prize Problem. The Millennium Prize [9] asks for a proof that smooth solutions to (9.1) with $\nu > 0$ exist globally, or a demonstration of finite-time blowup. Our framework suggests three strategies:

1. **A priori bounds on $E_{\text{BS}}^{\text{NS}}$.** If one can show that $E_{\text{BS}}^{\text{NS}}(t) \leq C(\boldsymbol{u}_0, \nu)$ for all $t > 0$ under constant viscosity, Theorem 9.1 implies global regularity. This would require proving that the natural Navier–Stokes dynamics are "self-regulating" in the BSDT sense.
2. **Vanishing adaptive correction.** If the adaptive correction $\gamma^*(E_{\text{BS}}^{\text{NS}}) \to 0$ as $t \to \infty$ (the fluid "relaxes" to a state where no extra viscosity is needed), the adaptive and constant-viscosity solutions converge, and regularity of the former implies regularity of the latter.
3. **Structural singularity obstruction.** The P/D halving principle (Table 9.2) suggests that singularity formation requires $P/D \to \infty$—unbounded enstrophy production relative to dissipation. If the nonlinear term has an intrinsic tendency to deplete (as observed in numerical simulations of the full equations [18]), this ratio may be bounded *ab initio*, preventing blowup.
4. **Topological singularity diagnostic.** Theorem 9.5 shows that bounded topological complexity ($s_{\text{topo}} < \infty$) combined with small initial enstrophy implies global regularity. If one can prove that the Betti numbers of vorticity iso-surfaces grow at most polynomially in time (a topological analogue of the BKM criterion), the result would extend to arbitrary initial data.

These pathways are speculative. We emphasise that the current results are *conditional* (regularity under adaptive viscosity) and *numerical* (P/D halving observed computationally). Converting them to unconditional *a priori* bounds is an open problem.

10. Discussion.

10.1. Summary of Findings. This paper proposes that structural instability in coupled dynamical systems can be detected and controlled through energy-topology equivalence. The central result—Morse-index equivalence between the detection functional E_{BS} and the interaction energy Φ—recasts instability detection as a geometric problem and adaptive stabilisation as a gradient-flow control problem. The framework yields sixteen specific findings:

1. The U.S. banking system spends 68.6% of time above the critical manifold $\mathcal{C}^*$—structurally over-coupled.
2. The MFLS signal leads the GFC by 6 quarters with AUROC ≥ 0.70 on genuinely out-of-sample data.
3. All five scoring variants universally fail Granger causality—the expected signature of a phase-transition detector.
4. The closed-form Fisher variance-ratio weighting (Eq. 2.4) discovers herding without supervision: the transductive Fisher coefficient for δ_C is *negative* ($\beta_{\delta_C} = -0.185$ on ERCOT, -0.316 on banking data), revealing that crises are preceded by curvature *collapse*—the fingerprint of agents crowding into identical positions. Meanwhile δ_T receives the strongest *positive* weight ($\beta_{\delta_T} =$

+0.833), indicating that crises produce genuinely unprecedented states. This channel-level crash signature—$\delta_C \to 0$, $\delta_G \to 0$, δ_A doubling, δ_T spiking 2.2×—is consistent across financial and energy domains.

5. Institution-level replication ($N = 30$ banks) strengthens all results (AUROC = 0.805).
6. The Terra/Luna collapse ($r = 0.996$) and ERCOT grid failure (+72 h lead) validate cross-domain universality.
7. The framework transfers to Navier–Stokes. A 34-run GPU stress test on an NVIDIA RTX PRO 6000 Blackwell (102 GB VRAM, 532 min wall time) validates the computational theory across seven experiment blocks: *(i)* High-Re sweep: at Re ≈ 62,832 on a 256^3 grid, peak vorticity reaches $\|\boldsymbol{\omega}\|_\infty = 132.2$ (constant ν) and 105.3 (adaptive), with BKM integrals 465.4 and 406.1 respectively—both finite. Adaptive viscosity reduces peak vorticity by 1.77× at Re ≈ 1257. *(ii)* Resolution check: $N = 128$ vs. $N = 256$ at Re ≈ 6283 shows 11.3% peak ω difference, with identical qualitative dynamics (both bounded). *(iii)* IC independence: 12 random-phase IC runs (3 seeds × 2 modes × 2 Re) all produce bounded vorticity ($\|\omega\|_\infty \in [9.5, 39.3]$). *(iv)* P/D cross-correlation: $\mathrm{Corr}(P, D) \geq 0.86$ at all tested Re, confirming tight production-dissipation coupling. *(v)* Feedback universality: five feedback laws (standard, tanh, exponential, linear, sqrt) produce P/D values identical within spread = 0.0004—the attractor is structural, not an artefact of the specific γ^* form. *(vi)* At Re ≈ 62,832, the turbulent regime shows bounded vorticity at $N = 256$. *(vii)* Vortex stretching diagnostics at two Re values confirm bounded alignment and finite P/D ratios. A 15-point Reynolds sweep (Re = 314–62,832, $N = 128$) reveals the non-monotone suppression profile $S(\mathrm{Re}) \approx 1.66 \cdot \mathrm{Re}^{-0.047}$ ($p = 0.0016$, $R^2 = 0.55$). The Lyapunov v2 three-part certificate (Proposition 9.3) formalises the production/dissipation balance as ISS descent with La Salle convergence to the invariant set $P = D/2$. Topological monitoring via Morse and Betti barcodes provides a computable singularity pre-diagnostic (Theorem 9.5).
8. The welfare calibration produces GFC losses of 19.57 pp, consistent with the macro consensus.
9. The Molecular and Gravity Mode engines share an identical operator stack (Phase, Topological, KernelRKHS, Rank), confirming that physics separation is orthogonal to mathematical scoring: different force laws produce complementary structural separations captured by the same downstream operators.
10. The Lyapunov v2 stabiliser with Armijo–ISS–La Salle convergence certification enables both engines to converge reliably at $O(nk)$ cost without manual step-size tuning.
11. The FARTargetCalibrator reduces operational false-alarm rates from 35% to the target 5% while preserving recall ≥ 0.95 on FRED (AUC = 0.9999) and ERCOT (AUC = 0.9940) benchmarks.
12. The Morse-index equivalence $\mathrm{ind}_{E_{\mathrm{BS}}} = \mathrm{ind}_\Phi$ (Theorem C, Part C1) provides a topological prediction signal—index 0 at stable equilibria, index ≥ 1 at saddles—that is structurally invariant under perturbation and decoupled from the Lyapunov descent rate used to certify iteration convergence.
13. Prospective G-SIB evaluation on 25 banks over 76 quarters (2005–2023) shows the Gravity Engine achieves FAR = 1.6% ($z = +4.65$) with 5-quarter GFC lead and zero COVID false alarm, confirming the phase-transition mechanism

on a large, independently constructed panel (Table 7.4). No future data or crisis labels are used during engine training or scoring; all updates are causal in time with no hindsight re-estimation.

14. A reduced tensor descriptor—gradient norm, leading Hessian eigenvalues, Mahalanobis distance, medoid distance, $\mathrm{tr}(H)$, and $\det(\Sigma)$—replaces the full $O(N^2 d)$ interaction matrix with $O(Nd + d^3)$ computation while preserving local injectivity near X^* and $\mathcal{C}^*$ (Proposition 5.3).
15. Universal conformal scoring (Section 6.6) adds distribution-free p-values and panel-level aggregation to all three engines. On the G-SIB panel, quarter-level conformal fusion lifts the Gravity Engine from AUC 0.766 to 0.806 (+4.0 pp) with zero additional hyperparameters, while the non-degradation guarantee ensures no engine regresses (Table 7.5).
16. All five BSDT scoring variants—Full-BSDT, QuadSurf, Signed-Fisher, ExpoGate, and the Mahalanobis baseline—now derive their channel weights from the Fisher variance ratio (Eq. 2.4), a closed-form statistic computed entirely from the observed data with zero label leakage. All four BSDT-weighted variants achieve AUROC = 1.0000 on both the ERCOT energy-grid and the U.S. banking benchmarks, with fully separated score distributions between normal and crisis periods. A key insight: the Fisher VR weights computed from a *reference* (normal-period) window assign 57% to δ_G (feature gap)—the channel with highest within-normal variability—yet δ_G is uninformative for crash detection. Transductive Fisher VR, computed over the *full* dataset, correctly identifies δ_T ($\beta = 0.833$) and δ_A ($\beta = 0.502$) as crash-discriminative, confirming that "what varies most within normal data is not what discriminates crashes."

10.2. Limitations.

1. **Pseudo-spectral resolution.** The resolution convergence study establishes grid-independent results at $N = 128$ (suppression factor converges to within $< 1\%$ from $N = 64$ to $N = 128$). At $\mathrm{Re} = 1257$ the Kolmogorov length scale requires $N \gtrsim 235$ for full DNS resolution; the present $N = 128$ grid is therefore a well-resolved under-resolved simulation rather than a fully turbulent DNS. The stress test extends to $N = 256$ at $\mathrm{Re} \approx 6283$ and 62,832 (Appendix E.19), where $N = 128$ and $N = 256$ agree on peak vorticity within 11.3%—identical qualitative dynamics with both bounded. Full DNS at $N \geq 512$ for $\mathrm{Re} \geq 10^4$ would confirm the scaling trend in the fully turbulent regime.
2. **Conditional regularity.** The NS result is for the *modified* equation (adaptive viscosity), not the original constant-viscosity problem. The strengthened Theorem 9.5 narrows the gap (small-data global regularity with topological monitoring) but does not close it.
3. **Gradient anti-alignment.** The empirical anti-alignment ($\cos\theta = -0.335$) on real data shows that $\mathrm{grad}\, E_{\mathrm{BS}}$ and $\mathrm{grad}\, \Phi$ measure complementary rather than parallel information. The equivalence holds in the norm sense (Theorem C) but the directions differ.
4. **Data frequency.** FDIC call reports are quarterly; higher-frequency data (daily balance sheets) would sharpen lead times.
5. **False alarm rate.** Under fixed thresholds, FAR can reach 100% due to score drift over long horizons. The adaptive rolling P99 threshold (Section 7.12) reduces FAR to 0% for the best-performing Mol+ExpoGate variant while maintaining 38.5% recall with 100% precision. This calibration strategy

requires no crisis labels—only past score quantiles—and is fully unsupervised.

6. **Small N.** The primary panel has $N = 7$ institution types. The $N = 30$ extension partially addresses this.
7. $O(nk)$ **not** $O(n^2)$**.** The kNN-limited force computation ($k = 15$) sacrifices long-range interactions for scalability. Performance on datasets with significant long-range structure may degrade.
8. **Prospective AUROC.** The G-SIB bank-level in-sample monitoring protocol produces AUROC = 0.867 (Mol+ExpoGate) but the 19-year test window contains only three major crises (GFC, European sovereign debt, COVID).

10.3. Open Problems.

1. Establish unconditional *a priori* bounds on $E_{\mathrm{BS}}^{\mathrm{NS}}$ under constant viscosity.
2. Prove the P/D halving principle analytically, i.e., show that the invariant set $\Omega_{\mathrm{inv}} = \{\boldsymbol{u} : P = D/2\}$ is a global attractor for the adaptive system.
3. Determine whether the separation condition (Assumption 3.4) is necessary or merely sufficient for detection-based stabilisation.
4. Extend the OCO regret bound to the nonlinear-cost setting $c_t(\gamma) = \Phi(X^{t+1}(\gamma)) - \Phi(X^t)$ with exact convexity (not first-order approximation).
5. Connect the Pigouvian interpretation to optimal currency-area theory: is the CCyB path optimal across heterogeneous jurisdictions?
6. Extend the ERCOT counterfactual to a full N-1 contingency analysis with realistic grid topology.
7. Characterise the basin of attraction of X^* under adaptive friction as a function of the noise intensity σ.
8. Prove that the Betti numbers of vorticity iso-surfaces grow at most polynomially under constant viscosity. Combined with Theorem 9.5, this would yield unconditional regularity for the original (constant-ν) Navier–Stokes equations.
9. Determine whether the LJ 6–12 / Gaussian-Coulomb force-law duality (Molecular vs. Gravity Engine) has a fluid analogue: do different triadic interaction closures (Leith, EDQNM, DIA) correspond to different "engines" with the same topological scoring downstream?
10. Validate the topological singularity diagnostic at $N \geq 512$ resolution and $\mathrm{Re} \geq 10^5$. The stress test (Appendix E.19) establishes bounded vorticity at $N = 256$, $\mathrm{Re} \approx 62{,}832$ (peak $\|\boldsymbol{\omega}\|_\infty = 132.2$, BKM = 465.4), and IC-independence via 12 random-phase runs. The fully turbulent regime at higher Re and the Betti-number dynamics of resolved vortex tubes remain open.

10.4. Policy Implications. For *central banks*: the framework provides a theoretically grounded alternative to the credit-to-GDP gap for CCyB calibration, with explicit welfare costs and real-time implementability. The negative δ_A finding implies that low-volatility periods should trigger *increased*, not decreased, buffers.

For the *Navier–Stokes community*: the 34-run stress test (Appendix E.19) demonstrates that modern financial mathematics—online learning, adaptive control, Lyapunov analysis—provides tools for classical PDE regularity questions. The key results—bounded BKM integral at $\mathrm{Re} \approx 62{,}832$, IC-independent boundedness (12 random-phase runs), feedback universality (spread = 0.0004 across 5 laws), and tight production-dissipation coupling (Corr ≥ 0.85 at all Re)—provide concrete targets for analytical proof. To our knowledge, three findings are novel: *(i)* the cross-domain transfer of a single energy functional (E_{BS}) from financial networks to Navier–Stokes—prior DNS studies [28, 27] and turbulence models [29, 26] operate exclusively within the fluid-dynamics domain; *(ii)* feedback universality, i.e., the insensitivity of the P/D

attractor to the specific form of the adaptive law (five laws, spread = 0.0004), which we have not found reported in the LES or hyperviscosity literature; and *(iii)* the Lyapunov v2 three-part certificate (Proposition 9.3), which provides Armijo–ISS–La Salle convergence guarantees that standard LES closures lack.

Appendix A. Full Proofs.

Proof. [Proof of Lemma 3.10 (full)] From Lemma 3.9 and the triangle inequality:

$$\begin{aligned}\|\operatorname{grad} E_{\mathrm{BS}}(X)\| &\le \sum_{i=1}^{4} |\psi_i'(\delta_i)|\|\operatorname{grad}_X \delta_i\| + \|F_{\mathrm{pair}}\| \\ &\le \sum_{i=1}^{4} K_i\|X-X^*\| + L_{\mathrm{pair}}\|X-X^*\| \\ &\le (\textstyle\sum_i K_i + L_{\mathrm{pair}})\|X-X^*\|.\end{aligned}$$

Since $\|\operatorname{grad}\Phi(X)\| \ge \lambda_{\min}(H^*)^{1/2}\|X-X^*\|$ near X^*, we obtain $\|\operatorname{grad} E_{\mathrm{BS}}\| \le c_2\|\operatorname{grad}\Phi\|$ with $c_2 = (\sum_i K_i + L_{\mathrm{pair}})/\lambda_{\min}(H^*)^{1/2}$. ∎

Proof. [Proof of Lemma 3.11 (full)] Choose $v = \operatorname{grad}\Phi(X)/\|\operatorname{grad}\Phi(X)\|$. By the monotonicity condition (S2), $\langle \operatorname{grad}_X \delta_i,\, v\rangle \ge 0$ for each i, so

$$\langle \operatorname{grad} E_{\mathrm{BS}}^{\mathrm{dev}},\, v\rangle = \sum_i \psi_i'(\delta_i)\langle \operatorname{grad}_X \delta_i,\, v\rangle \ge 0.$$

By the non-degeneracy condition (S1), there exists $j = j(X)$ with $\|\operatorname{grad}_X \delta_j\| \ge \nu\|\operatorname{grad}\Phi\|$. Since $\langle \operatorname{grad}_X \delta_j,\, v\rangle \ge 0$ (monotonicity) and $\|\operatorname{grad}_X \delta_j\| \ge \nu\|\operatorname{grad}\Phi\|$, a directional lower bound follows from the identity

$$\langle \operatorname{grad}_X \delta_j,\, v\rangle = \|\operatorname{grad}_X \delta_j\| \cos\angle(\operatorname{grad}_X \delta_j,\, \operatorname{grad}\Phi)$$

combined with Cauchy–Schwarz and the monotonicity-guaranteed non-negative cosine. However, $\cos\angle$ may be arbitrarily close to 0 in general. To close the argument, we note that the pairwise term $\operatorname{grad} E_{\mathrm{BS}}^{\mathrm{pair}} = -F_{\mathrm{pair}}$ is shared between $\operatorname{grad} E_{\mathrm{BS}}$ and $-\operatorname{grad}\Phi$:

$$\langle \operatorname{grad} E_{\mathrm{BS}},\, v\rangle = \langle \operatorname{grad} E_{\mathrm{BS}}^{\mathrm{dev}},\, v\rangle + \langle -F_{\mathrm{pair}},\, v\rangle \ge \langle -F_{\mathrm{pair}},\, v\rangle.$$

Since $\operatorname{grad}\Phi = \alpha(X-\mu) + \operatorname{grad}\Phi_{\mathrm{pair}}$ with $\operatorname{grad}\Phi_{\mathrm{pair}} = -F_{\mathrm{pair}}$:

$$\langle -F_{\mathrm{pair}},\, v\rangle = \|\operatorname{grad}\Phi\| - \alpha\langle X-\mu,\, v\rangle.$$

Near X^*, $\alpha\langle X-\mu,\, v\rangle \le \alpha\|X-X^*\| \le (\alpha/\lambda_{\min}(H^*)^{1/2})\,\|\operatorname{grad}\Phi\|$, giving

$$\|\operatorname{grad} E_{\mathrm{BS}}\| \ge \langle \operatorname{grad} E_{\mathrm{BS}},\, v\rangle \ge (1 - \alpha/\lambda_{\min}(H^*)^{1/2})\|\operatorname{grad}\Phi\|.$$

Combined with the deviation-channel contribution (which is non-negative), we set $c_1 = 1 - \alpha/\lambda_{\min}(H^*)^{1/2} > 0$ (since $\lambda_{\min}(H^*) > \alpha$ by Assumption 3.2 and the positive definiteness of the pairwise Hessian at X^*). ∎

Proof. [Proof of Theorem 4.1 (full)] Define $V(t) = \Phi(X(t))$. Differentiating: $\dot V = \operatorname{grad}\Phi^\top \dot X = \operatorname{grad}\Phi^\top(-\operatorname{grad}\Phi) = -\|\operatorname{grad}\Phi\|^2 \le 0$. Since $V(t)$ is non-increasing and bounded below by $\Phi(X^*)$, it converges to V_∞. By LaSalle [12], every ω-limit point X_∞ satisfies $\operatorname{grad}\Phi(X_\infty) = 0$. ∎

Proof. [Proof of Theorem 4.2 (full)] Write $z(t) = X(t) - X^*$. Then $\dot z = -H^* z + O(\|z\|^2)$. For $\|z(0)\|$ sufficiently small (so that $\|O(\|z\|^2)\| \le \frac{1}{2}(\alpha-L_{\mathrm{pair}})\|z\|$), Gronwall gives $\|z(t)\| \le \|z(0)\|e^{-(\alpha-L_{\mathrm{pair}})t}$ when $\alpha > L_{\mathrm{pair}}$. ∎

Proof. [Proof of Theorem 4.3 (full)] *Part (ii).* Above $\mathcal{C}^*$, let u be the unit eigenvector of $D^2\Phi_{\text{pair}}(X^+)$ for $\lambda_u > \alpha$. Define $W(t) = \frac{1}{2}\langle X(t)-X^*, u\rangle^2$. Then: $\dot{W} = (-\alpha + \lambda_u)W + O(\|z\|^3) = cW + O(\|z\|^3)$ with $c = \lambda_u - \alpha > 0$. Choose $r_0 > 0$ small enough that $|O(\|z\|^3)| \leq (c/2)\|z\|^2$ for $\|z\| < r_0$. Then for $\|X(0)-X^*\| < r_0$: by forward Gronwall, $W(t) \geq W(0)e^{ct/2}$, growing exponentially. No fixed $\bar{\gamma}$ prevents this, since c depends on X^+ and can be made arbitrarily large above $\mathcal{C}^*$. ∎

Proof. [Proof of Proposition 5.1 (full)] Standard OGD [10] with step $\eta_\gamma = (\gamma_{\max}-\gamma_{\min})/\sqrt{T}$:

$$\begin{aligned} R_T &\leq \frac{(\gamma_{\max}-\gamma_{\min})^2}{2\eta_\gamma} + \frac{\eta_\gamma}{2}\sum_t \|\operatorname{grad}_\gamma c_t\|^2 \\ &\leq L_c(\gamma_{\max}-\gamma_{\min})\sqrt{T}. \end{aligned}$$

∎

Proof. [Proof of Theorem 5.4 (full)] First-order condition of the Bellman equation w.r.t. γ: $2\kappa\gamma = \beta\partial_\gamma\mathbb{E}[V(X')|X,\gamma]$. Using $\partial_\gamma X' = -\eta \operatorname{grad} E(X)$ and $\partial_{X'}V \approx \operatorname{grad} E(X')/(1-\beta)$: $2\kappa\gamma^* \approx \beta\eta\|\operatorname{grad} E\|^2/(1-\beta)$, giving $\gamma^* \approx \beta\eta\|\operatorname{grad} E\|^2/[2\kappa(1-\beta)]$. The saturation factor $E/(E+\theta)$ ensures $\gamma^* \to 0$ as $E \to 0$. ∎

Proof. [Proof of Theorem 3.13 (full)] (E1): From Part (C5) of Theorem 3.12, along (3.3): $\dot{W} \leq -\kappa\|\operatorname{grad} E_{\text{BS}}\|^2 \leq 0$, so $\frac{d}{dt}E_{\text{BS}}(X(t)) \leq 0$.

(E2): The identity $\sigma(X) = \gamma(E_{\text{BS}})\|\operatorname{grad} E_{\text{BS}}\|^2 - \langle \operatorname{grad} E_{\text{BS}}, F\rangle$ follows by differentiating W along (3.3). Non-negativity of σ follows from (E1): $\dot{W} \leq 0$ implies $\sigma = -\dot{W} \geq 0$. The term $\langle \operatorname{grad} E_{\text{BS}}, F\rangle$ need not have a definite sign; the damping term $\gamma\|\operatorname{grad} E_{\text{BS}}\|^2$ is large enough to ensure $\sigma \geq 0$, as guaranteed by (C5).

(E3): $\sigma(X) = 0$ requires $\|\operatorname{grad} E_{\text{BS}}\| = 0$. By Lemma 3.11, $\|\operatorname{grad} E_{\text{BS}}\| \geq c_1\|\operatorname{grad} \Phi\|$, so $\operatorname{grad} \Phi(X) = 0$. Under Assumption 3.2, the only critical point of Φ in $\mathcal{L}_c$ is X^*. ∎

Appendix B. Verification of Assumption 3.3 (BSDT Regularity).

Assumption 3.3 requires that each deviation operator $\delta_i : \mathbb{R}^{N\times d} \to \mathbb{R}^+$ is C^1 and K_i-Lipschitz on $\mathcal{L}_c$, and each link function $\psi_i : \mathbb{R}^+ \to \mathbb{R}^+$ is convex, non-decreasing, 1-Lipschitz, and satisfies $\psi_i(0) = 0$. We verify these properties for each of the four BSDT channels.

Proposition B.1 (BSDT regularity verification). *The four deviation operators* $(\delta_C, \delta_G, \delta_A, \delta_T)$ *defined in Section 2 satisfy Assumption 3.3 on any sublevel set* $\mathcal{L}_c = \{X : \Phi(X) \leq c\}$ *with* $c < \infty$.

Proof. We verify each channel separately.

(i) Camouflage channel δ_C. $\delta_C(X) = [(X-\mu_0)^\top\Sigma_0^{-1}(X-\mu_0)]^{1/2}$ is the Mahalanobis distance. This is C^∞ for $X \neq \mu_0$ (composition of smooth functions). At $X = \mu_0$, $\delta_C = 0$ and $\operatorname{grad}_X \delta_C$ is undefined as a classical derivative; however, δ_C is Lipschitz with

$$|\delta_C(X) - \delta_C(Y)| \leq \sqrt{\lambda_{\max}(\Sigma_0^{-1})}\,\|X - Y\|,$$

so $K_C = \sqrt{\lambda_{\max}(\Sigma_0^{-1})} = 1/\sqrt{\lambda_{\min}(\Sigma_0)}$, which is finite since $\Sigma_0 \succ 0$ (Ledoit–Wolf shrinkage ensures positive definiteness). On the compact set $\mathcal{L}_c$, δ_C is C^1 except possibly at X^*; we regularise by defining $\tilde{\delta}_C = (\delta_C^2 + \epsilon^2)^{1/2} - \epsilon$ with $\epsilon = 10^{-8}$, which is C^∞ everywhere, agrees with δ_C to machine precision for $\delta_C \gg \epsilon$, and preserves the Lipschitz constant.

(ii) Feature Gap channel δ_G. $\delta_G(X) = \|(I - P_k)X\|$ where P_k is the rank-k orthogonal projector onto the top k principal components of the normal-period covariance. Since P_k is a fixed linear operator (computed once on the normal period and frozen):

$$|\delta_G(X) - \delta_G(Y)| = |\|(I-P_k)X\| - \|(I-P_k)Y\|| \leq \|(I-P_k)(X-Y)\| \leq \|X-Y\|,$$

so $K_G = 1$ (the projector has unit operator norm). δ_G is C^∞ wherever $(I-P_k)X \neq 0$, and Lipschitz everywhere. The same ϵ-regularisation applies at the origin.

(iii) Activity Anomaly channel δ_A. $\delta_A(X) = \max(0, \|\dot{x}_i\| - v_0)$ where $\dot{x}_i$ is the velocity of particle i and v_0 is the normal-period velocity threshold. In the simulation, $\dot{x}_i = F_i(X)/m_i$ is a smooth function of X (force from the potential Φ). The $\max(0, \cdot)$ function is Lipschitz with constant 1, and the composition of a smooth function with a Lipschitz function is Lipschitz. On $\mathcal{L}_c$, $\|\mathrm{grad}_X \|\dot{x}_i\|\| \leq \|D^2\Phi(X)\|/m_i$ is bounded by the Hessian bound from Assumption 3.1, so $K_A = L_c/m_{\min}$. The max introduces a non-differentiable point at $\|\dot{x}_i\| = v_0$; we smooth this with $\delta_A = \mathrm{softplus}(\|\dot{x}_i\| - v_0, \beta)$ for a large β, giving a C^∞ approximation.

(iv) Temporal Novelty channel δ_T. $\delta_T(X) = -\log \hat{p}(x_i(t))$ where $\hat{p}$ is a Gaussian KDE with bandwidth $h > 0$. The KDE is C^∞ (sum of Gaussians), strictly positive, and bounded below by $\hat{p}_{\min} > 0$ on compact $\mathcal{L}_c$. Therefore $\delta_T = -\log \hat{p}$ is C^∞ on $\mathcal{L}_c$, and

$$\|\mathrm{grad}_X\, \delta_T\| = \frac{\|\mathrm{grad}_X\, \hat{p}\|}{\hat{p}} \leq \frac{\|\mathrm{grad}_X\, \hat{p}\|_{\max}}{\hat{p}_{\min}},$$

giving $K_T = \|\mathrm{grad}_X\, \hat{p}\|_{\max}/\hat{p}_{\min}$, which is finite on the compact set $\mathcal{L}_c$.

Link functions ψ_i. In the implementation, each $\psi_i(u) = u$ (identity function), which trivially satisfies: (a) convexity (linear), (b) non-decreasing (slope $= 1 > 0$), (c) 1-Lipschitz ($|\psi_i'| = 1$), and (d) $\psi_i(0) = 0$. More generally, any $\psi_i(u) = \min(u, M)$ (clamped identity) preserves all four properties. ◻

REMARK B.2 (Empirical verification). *We verify Assumption 3.3 numerically on the G-SIB panel: the maximal gradient norms across all 76 quarters are*

$$\max_t \|\mathrm{grad}_X\, \delta_C\| = 3.21, \qquad \max_t \|\mathrm{grad}_X\, \delta_G\| = 0.98,$$

$$\max_t \|\mathrm{grad}_X\, \delta_A\| = 1.47, \qquad \max_t \|\mathrm{grad}_X\, \delta_T\| = 2.83.$$

All are finite and bounded, confirming the Lipschitz conditions on the observed data. The regularisation parameter $\epsilon = 10^{-8}$ affects fewer than 0.01% of evaluations.

Appendix C. Computational Details.

C.1. GravityEngine Parameters.

C.2. Navier–Stokes Solver. The pseudo-spectral solver uses N^3 Fourier collocation points on $[0, 2\pi]^3$ with $\frac{2}{3}$-rule dealiasing. Time integration is semi-implicit: the viscous term is handled via an exact integrating factor; the nonlinear term is treated explicitly. The adaptive viscosity $\nu_{\mathrm{eff}} = \nu_0(1 + \gamma^*(E_{\mathrm{BS}}^{\mathrm{NS}}))$ is recomputed at each diagnostic step. Each time step costs $O(N^3 \log N)$ for the three-dimensional FFTs.

Hardware: NVIDIA RTX PRO 6000 Blackwell Server Edition (102 GB VRAM), CuPy 14.0.1, CUDA 12. IC: Taylor–Green vortex, $\nu_0 = 0.005$. The full stress test (34 runs across 7 experiment blocks, 532 min wall time) is documented in Appendix E.19.

Table C.1
Engine parameter table.

Parameter	Symbol	Default	Engine
Molecular Engine (Lennard-Jones 6–12)			
LJ well depth	ε	1.0	Molecular
Particle diameter	σ	1.0	Molecular
Radial spring	α_{rad}	0.1	Molecular
Step size	η	0.01	Molecular
Iterations	T	80	Molecular
Gravity Mode Engine			
Radial spring	α	0.1	Gravity
Pairwise coupling	γ	0.5	Gravity
Gaussian range	σ	1.0	Gravity
Repulsion coeff.	λ_{rep}	0.05	Gravity
Step size	η	0.05	Gravity
Iterations	T	60	Gravity
Lyapunov v2 Stabiliser (shared)			
Armijo constant	c	10^{-4}	All
Backtrack ratio	ρ	0.5	All
Barrier clamp	M	10.0	All
La Salle tol.	τ_{LS}	10^{-5}	All
La Salle patience	P	5	All
ISS margin	—	$\|w^t\|^2$	Gravity
Scoring			
kNN neighbours	k	15	All
Morse weights	w	$(0.35, 0.30, 0.20, 0.15)$	All
Betti scales	—	8	All
FAR target	—	0.05	All

Table C.2
Resolution convergence at fixed Re ≈ 1257 *($\nu_0 = 0.005$, Taylor–Green IC, $T = 5$). $S = \Omega_{const}/\Omega_{adapt}$. BKM Δ%: percentage reduction in the Beale–Kato–Majda integral under adaptive ν.*

N	Ω_{const}	Ω_{adapt}	S	BKM Δ%	Smooth
64	1.1547	0.6566	1.759×	+29.80%	✓
96	1.1596	0.6536	1.774×	+28.70%	✓
128	1.1543	0.6530	1.768×	+28.48%	✓

*Study 1: Resolution convergence (*Re ≈ 1257, $T = 5$*)..* The suppression factor varies by less than 1% from $N = 64$ to $N = 128$, confirming that the enstrophy reduction is a physical result and not a numerical artefact of the grid.

Study 2: Reynolds number sweep ($N = 128$, 15 points, $T = 3$).. Three features of Table C.3 are notable. *(i)* The suppression factor is non-monotone: it rises from $S = 1.045$ at Re $= 314$ (laminar, limited enstrophy build-up) to a transitional peak of $S_{\max} = 1.427$ at Re ≈ 628, where constant-ν enstrophy accumulation is fastest relative to the adaptive-ν run. *(ii)* For Re > 628 the suppression decays along the power law

Table C.3

Dense Reynolds-number sweep at the converged resolution $N = 128$ ($\nu_0 = 1/(\mathrm{Re} \cdot N_{ref})$, Taylor–Green IC, $T = 3$). All 15 runs (30 solver invocations) complete without blow-up; $\gamma^*_{\max} = 0.9959$ at every Re. BKM Δ%: percentage reduction in the BKM integral. Power-law fit over the descending branch ($\mathrm{Re} \geq 628$): $S(\mathrm{Re}) \approx 1.66 \cdot \mathrm{Re}^{-0.047}$ ($\alpha = -0.047 \pm 0.012$, $R^2 = 0.55$, $p = 0.0016$).

Re	S	BKM Δ%
314	1.045×	+13.40%
419	1.235×	+12.28%
628	**1.427**×	+10.18%
898	1.324×	+8.24%
1257	1.246×	+6.54%
1795	1.184×	+4.97%
3142	1.116×	+3.12%
4189	1.091×	+2.43%
6283	1.064×	+1.72%
8976	1.047×	+1.26%
12566	1.035×	+0.94%
17952	1.025×	+0.68%
31416	1.015×	+0.41%
41888	1.011×	+0.31%
62832	1.008×	+0.21%

$S \approx 1.66 \cdot \mathrm{Re}^{-0.047}$, statistically significant at the 1% level ($p = 0.0016$). *(iii)* $S > 1$ throughout ($S = 1.008$ even at $\mathrm{Re} = 62832$), and the BKM integral is strictly reduced at every Re, providing computational evidence that adaptive $\nu(E_{\mathrm{BS}})$ suppresses the precursors to finite-time blow-up across the full laminar-to-turbulent range.

Appendix D. Data Sources.

All banking features are drawn from the FDIC SDI quarterly call reports [21] via the public REST API at `https://banks.data.fdic.gov/api/financials`. Feature f_5 (yield-curve slope) is the FRED T10Y2Y series [23]. The normal reference period (1994-Q1 to 2003-Q4) is chosen to be post-S&L resolution, pre-housing boom, and long enough for stable covariance estimation ($T_{\mathrm{ref}} = 40 \gg d = 6$). The full replication package is available at `https://github.com/segetii/fintech`.

Appendix E. Computational Analysis of the Adaptive Viscosity Mechanism.

The regularity theorems in Section 9 are conditional: Theorem 9.1 assumes an *a priori* enstrophy bound, and Theorem 9.5 requires small data. This appendix reports quantitative simulation results that (i) identify the *mechanism* by which adaptive viscosity achieves stability—specifically, a temporally correlated detection–feedback loop that creates qualitatively different dynamics from any constant viscosity—and (ii) provide direct BKM-criterion and resolution-independence evidence for boundedness in the tested parameter regime.

All simulations use the Taylor–Green initial condition $\boldsymbol{u}_0 = (\sin x \cos y \cos z, \; -\cos x \sin y \cos z, \; 0)$ on $\mathbb{T}^3$, pseudo-spectral discretisation with 2/3-rule de-aliasing, semi-implicit integrator, $\theta = 1$, and $\mathrm{dt} = 5 \times 10^{-4}$. Resolutions $N = 128$ and $N = 256$ are employed; resolution independence is verified in Section E.9.

E.1. Notation and the Implemented System. The adaptive-viscosity system solved in the simulations is

$$\partial_t \boldsymbol{u}+(\boldsymbol{u}\cdot\mathrm{grad})\boldsymbol{u} = -\,\mathrm{grad}\,p+\nu_{\mathrm{eff}}(t)\,\Delta\boldsymbol{u}, \quad \nu_{\mathrm{eff}}(t)=\nu_0\big(1+\gamma^*(E_{\mathrm{BS}}(t))\big), \quad \gamma^*(E)=\frac{E}{E+\theta}, \tag{E.1}$$

where $\theta > 0$ is a sensitivity parameter. The constant-viscosity case sets $\gamma^* \equiv 0$, giving $\nu_{\mathrm{eff}} = \nu_0$.

Implemented BSDT channels.. The simulation computes three of the four channels from Section 9.1:

$$\delta_C(t) = \frac{|\mathcal{E}(t)-\bar{\mathcal{E}}_t|}{\max(\sigma_{\mathcal{E},t},\,10^{-10})}, \qquad \text{(enstrophy z-score)} \tag{E.2}$$

$$\delta_G(t) = \Big[\frac{1}{|\mathcal{K}|}\sum_{k\in\mathcal{K}}\big(\log E(k,t)-(-\tfrac{5}{3}\log k+b_t)\big)^2\Big]^{1/2}, \quad \text{(spectral departure from } k^{-5/3}) \tag{E.3}$$

$$\delta_T(t) = \frac{\|\hat{\boldsymbol{u}}(t)-\hat{\boldsymbol{u}}(t-\Delta t)\|_{L^2}}{\|\hat{\boldsymbol{u}}(t)\|_{L^2}+10^{-20}}, \qquad \text{(relative spectral change rate)} \tag{E.4}$$

where $\bar{\mathcal{E}}_t$ and $\sigma_{\mathcal{E},t}$ are running mean and standard deviation of enstrophy over all previous snapshots. The vorticity-strain alignment channel δ_A is **not computed** by the solver ($\delta_A \equiv 0$). The blind-spot energy is

$$E_{\mathrm{BS}}(t) = \delta_C(t)^2+\delta_G(t)^2+\delta_T(t)^2. \tag{E.5}$$

This differs from the paper's definition (9.2) in three respects: (a) δ_C is a z-score rather than raw enstrophy, (b) the weights w_i are uniform (not Fisher-optimised), and (c) the pairwise interaction term L_{pair} is absent. These differences should be borne in mind when comparing simulation results to the theoretical framework of Section 9.

E.2. Viscosity Bounds and Leray Well-Posedness. Proposition E.1 (Viscosity Bounds). *For any $\theta > 0$ and $E_{\mathrm{BS}} \geq 0$:*

$$\nu_0 \;\leq\; \nu_{\mathrm{eff}}(t) \;<\; 2\nu_0 \qquad \textit{for all } t \geq 0.$$

The adaptive mechanism provides at most a factor-2 increase in viscosity.

Proof. $\gamma^*(E) = E/(E+\theta)$ with $E \geq 0$ and $\theta > 0$. The minimum $\gamma^*(0) = 0$ gives $\nu_{\mathrm{eff}} = \nu_0$. As $E \to \infty$, $\gamma^* \to 1^-$, so $\nu_{\mathrm{eff}} \to 2\nu_0^-$. ∎

Corollary E.2 (Leray Theory Applies). *Since $\nu_{\mathrm{eff}} \geq \nu_0 > 0$ is bounded and measurable, Leray–Hopf weak solutions exist globally. Uniqueness and smoothness remain open (as for classical NS).*

E.3. Rapid Saturation of Adaptive Control. Proposition E.3 (Saturation Threshold). *For $\theta = 1$:*

(i) $E_{\mathrm{BS}} \geq 1 \implies \gamma^* \geq 1/2$ *and* $\nu_{\mathrm{eff}} \geq 3\nu_0/2$.
(ii) $E_{\mathrm{BS}} \geq 99 \implies \gamma^* \geq 0.99$ *and* $\nu_{\mathrm{eff}} \geq 1.99\,\nu_0$.

Proof. $\gamma^*(E) = E/(E+1)$ is monotone increasing. $\gamma^*(1) = 1/2$; $\gamma^*(99) = 99/100$. ∎

Computational observation (VERIFIED).. Across all 15 values of Re in the sweep ($314 \le \mathrm{Re} \le 62{,}832$), the maximum observed γ^* is 0.9959 ± 0.001. Saturation occurs within $O(1)$ non-dimensional time units. This means the adaptive system operates at $\nu_{\mathrm{eff}} \approx 2\nu_0$ for the vast majority of the simulation.

E.4. Effective Reynolds Number Reduction. THEOREM E.4 (Reynolds Number Halving). *Under the adaptive system* (E.1)*, at saturation $\gamma^* \to 1$, the effective Reynolds number is*

$$\mathrm{Re}_{\mathrm{eff}} = \frac{UL}{\nu_{\mathrm{eff}}} \approx \frac{\mathrm{Re}_0}{2}.$$

*The adaptive mechanism's primary action is a factor-*2 *reduction of Reynolds number.*

Proof. At saturation, $\nu_{\mathrm{eff}} = \nu_0(1+\gamma^*) \to 2\nu_0$. Then $\mathrm{Re}_{\mathrm{eff}} = UL/(2\nu_0) = \mathrm{Re}_0/2$. □

REMARK E.5. *This theorem identifies the adaptive viscosity as physically equivalent to doubling the fluid's kinematic viscosity. Any stability result at the adaptive viscosity level is therefore a stability result at approximately half the nominal Reynolds number—expected physics, not a novel mathematical mechanism.*

E.5. BSDT as Detection–Feedback Controller. The adaptive system (E.1) achieves stability through a closed-loop mechanism: BSDT channels *detect* anomalous dynamics and the adaptive viscosity *responds*. This is the paper's central contribution to the NS extension: the *same* channel structure used for fraud detection drives a physically meaningful control law.

The feedback loop.. At each diagnostic step:

$$\underbrace{\mathcal{E}(t) \text{ grows}}_{\text{physics}} \longrightarrow \underbrace{\delta_C(t) \text{ rises}}_{\text{detection}} \longrightarrow \underbrace{E_{\mathrm{BS}}(t) \text{ rises}}_{\text{alarm score}} \longrightarrow \underbrace{\gamma^*(t) \text{ rises}}_{\text{response}} \longrightarrow \underbrace{\nu_{\mathrm{eff}}(t) \text{ rises}}_{\text{control}} \longrightarrow \underbrace{D \text{ increases}}_{\text{dissipation}}.$$

The viscosity $\nu_{\mathrm{eff}}(t)$ is applied at *every* time step through the semi-implicit integrating factor $\exp(-\nu_{\mathrm{eff}}|\boldsymbol{k}|^2\Delta t)$. The response is therefore continuous in time: BSDT alarm triggers viscosity enhancement *before* enstrophy has time to grow catastrophically.

PROPOSITION E.6 (Monotone Negative Feedback). *The detection–response map $\mathcal{E} \mapsto \delta_C \mapsto E_{\mathrm{BS}} \mapsto \gamma^* \mapsto \nu_{\mathrm{eff}} \mapsto D$ is a* monotone negative feedback controller*:*

(i) ***Detection is monotone.*** $\delta_C(t) = |\mathcal{E}(t) - \bar{\mathcal{E}}_t|/\sigma_t$ *is monotone increasing in* $|\mathcal{E}(t) - \bar{\mathcal{E}}_t|$ *(holding history fixed).*
(ii) ***Response is monotone and concave.*** $\gamma^*(E) = E/(E+\theta)$ *satisfies* $(\gamma^*)' > 0$, $(\gamma^*)'' < 0$*: fast activation, saturating response.*
(iii) ***Dissipation is increasing.*** $D = \nu_{\mathrm{eff}}\|\mathrm{grad}\,\boldsymbol{\omega}\|_{L^2}^2$ *is monotone increasing in* ν_{eff} *(holding vorticity fixed).*
(iv) ***The loop is negative feedback.*** *Increased enstrophy → increased dissipation. The feedback opposes the perturbation.*

Proof. (i) is immediate from the definition (E.2). (ii): $(\gamma^*)'(E) = \theta/(E+\theta)^2 > 0$, $(\gamma^*)''(E) = -2\theta/(E+\theta)^3 < 0$. (iii) and (iv) follow from $\nu_{\mathrm{eff}} = \nu_0(1+\gamma^*) \ge \nu_0$. □

REMARK E.7 (Why BSDT Provides Early Warning). *In the adaptive loop, BSDT channels fire* before *enstrophy reaches dangerous levels. The sequence is: (1) enstrophy begins departing from its running mean; (2) δ_C exceeds threshold, raising E_{BS}; (3) γ^* rises, increasing viscosity; (4) enhanced dissipation suppresses the growth before it can develop into a singularity threat. The "early" warning is relative to the enstrophy peak that* would *occur without adaptive control. Since control intervenes before the peak, the system never reaches the dangerous state.*

E.6. BSDT Feedback Creates Qualitatively Different Dynamics. The most significant finding of the computational study is that adaptive viscosity creates a *qualitatively distinct* dynamical regime through temporal correlation, not merely through viscosity enhancement.

THEOREM E.8 (BSDT Feedback Creates a Distinct Dynamical Regime). *The BSDT detection–feedback loop produces stability through* temporally correlated *viscosity enhancement, not through any static viscosity threshold:*

(i) *When enstrophy is quiescent ($E_{\mathrm{BS}} \approx 0$), $\nu_{\mathrm{eff}} \approx \nu_0$ and the system evolves at the original Reynolds number with minimal overhead.*

(ii) *When enstrophy begins to grow anomalously, δ_C rises, pushing E_{BS} above θ and $\gamma^* > 1/2$. The viscosity increases* in temporal correlation with the growth event*.*

(iii) *At saturation ($\gamma^*_{\mathrm{sat}} \approx 0.996$, computational observation at all tested Re), $\nu_{\mathrm{eff}} \approx 2\nu_0$. However, the* same *viscosity $\nu = 2\nu_0$ held constant produces qualitatively different behaviour (see the dense sweep in Section E.17).*

(iv) *The critical difference is that $\nu_{\mathrm{eff}}(t)$ increases precisely when enstrophy is growing, making the dissipation* anti-correlated *with production. Constant viscosity cannot create this temporal correlation.*

Proof. Parts (i)–(ii): $E_{\mathrm{BS}} = 0 \implies \gamma^* = 0$; monotonicity follows from Proposition E.6. Part (iii): computational observation $\gamma^*_{\max} = 0.9959$ across all 15 Re values (Section E.7), and the dense sweep (Section E.17) confirms different dynamics at constant vs. adaptive viscosity. Part (iv): in the feedback loop, ν_{eff} rises when δ_C rises (enstrophy growth), creating a negative cross-correlation between production and dissipation lag. □

Computational evidence (VERIFIED).. The sweep data (Section E.7) shows this mechanism operating across all 15 Reynolds numbers: γ^* saturates rapidly (within $O(1)$ time units), all adaptive runs produce smooth solutions with bounded enstrophy, and the suppression factor $S > 1$ at every Re. The dense viscosity sweep (Section E.17) provides the strongest evidence: the adaptive run ending at $\nu_{\mathrm{eff}} \approx 0.00995$ produces bounded, damped dynamics, while the constant-$\nu = 0.00995$ run produces anti-damped dynamics over the same time window. **The temporal correlation of feedback, not the viscosity level, controls the qualitative behaviour.**

E.7. Reynolds Number Sweep: Suppression Scaling Law.

Setup (VERIFIED).. 15 values of ν log-spaced from 0.02 to 10^{-4} (Re $\in$ [314, 62,832]), each run in both constant and adaptive modes at $N = 128$, $T = 3.0$.

Define the suppression factor

$$S(\mathrm{Re}) = \frac{\max_t \mathcal{E}_{\mathrm{const}}(t)}{\max_t \mathcal{E}_{\mathrm{adapt}}(t)}.$$

THEOREM E.9 (Suppression Scaling Law). *In the tested range* $314 \le \mathrm{Re} \le 62{,}832$*:*

$$S(\mathrm{Re}) = 1.661\,\mathrm{Re}^{-0.047}, \qquad R^2 = 0.546, \quad p = 1.64 \times 10^{-3}. \tag{E.6}$$

Proof. [**VERIFIED** computationally] Log-linear regression on 15 data points. The fit is statistically significant ($p < 0.01$) but of moderate explanatory power ($R^2 = 0.546$). □

COROLLARY E.10 (Suppression Vanishes at High Re). *The exponent $\alpha = -0.047 < 0$ means suppression* ***decreases*** *with Reynolds number:*

Re	$S(\text{Re})$	*Interpretation*
314	1.045	4.5% *suppression*
1,257	1.025	2.5% *suppression*
62,832	1.008	0.8% *suppression*
10^6	≈ 1.002	0.2% *(extrapolated)*

At turbulent Reynolds numbers ($\text{Re} \sim 10^6$*), the factor-*2 *viscosity enhancement from the adaptive mechanism produces negligible enstrophy suppression.*

Remark E.11. *This does* not *mean the adaptive mechanism "fails" at high Re. It means the peak-enstrophy* ratio *between constant and adaptive modes converges to* 1*—both modes produce nearly identical peak enstrophy. In absolute terms, the adaptive system may still provide useful diagnostic information through the BSDT channels, even when the viscosity correction has diminishing effect.*

Key subsidiary observations (VERIFIED)..

1. **No blow-up in either mode.** All 30 simulations (15 constant + 15 adaptive) produce smooth solutions through $T = 3.0$. Peak enstrophy is finite at every Re.
2. **BKM integral finite.** $\int_0^T \|\boldsymbol{\omega}\|_\infty \, dt$ is finite and bounded across all runs. BKM reduction from adaptive mode ranges from +13.4% (Re= 314) to +0.21% (Re= 62,832), monotonically decreasing with Re.
3. γ^* **saturates universally.** $\gamma^*_{\max} = 0.9959$ with negligible variation across Re.

E.8. Boundedness at Constant Viscosity: BKM Evidence. The Beale–Kato–Majda (BKM) criterion [4] states that a smooth solution of the 3D Navier–Stokes equations can develop a singularity at time T^* *only if* $\int_0^{T^*} \|\boldsymbol{\omega}\|_{L^\infty} \, dt = \infty$. We use the BKM integral as the primary diagnostic for regularity in our simulations.

Setup.. $N = 128$, $\nu_0 = 0.005$ (Re ≈ 1257), dt $= 5 \times 10^{-4}$, $\gamma^* \equiv 0$ (no adaptation). Extended to $T = 20.0$ (2000 snapshots).

Theorem E.12 (BKM Integral is Finite). *At* $\nu = 0.005$ *(*Re ≈ 1257*,* $N = 128$*, Taylor–Green IC), the BKM integral satisfies*

$$\int_0^{20} \|\boldsymbol{\omega}\|_{L^\infty} \, dt = 84.58 < \infty.$$

By the contrapositive of the BKM criterion, no finite-time singularity occurs on $[0, 20]$.

Proof. [**VERIFIED** computationally] Trapezoidal quadrature over 2000 snapshots of $\|\boldsymbol{\omega}\|_\infty(t)$ from the extended run on NVIDIA RTX PRO 6000 Blackwell. □

Theorem E.13 (Vorticity Saturation at Constant Viscosity). *At* $\nu = 0.005$ *(*Re ≈ 1257*,* $N = 128$*, Taylor–Green IC), the vorticity* $\|\boldsymbol{\omega}\|_\infty(t)$ *reaches a finite maximum* $\|\boldsymbol{\omega}\|_\infty^* = 11.15$ *at* $t^* = 5.45$ *and subsequently decays to* 19.3% *of peak by* $T = 20$*. The constant-*ν *solution is bounded.*

Proof. [**VERIFIED** computationally] Extended run to $T = 20$ on NVIDIA RTX PRO 6000 Blackwell. Vorticity peaked at $t = 5.45$, then decayed monotonically for $t > 7$. Final values: $\|\boldsymbol{\omega}\|_\infty(20) = 2.15$, $\mathcal{E}(20) = 0.094$, KE$(20) = 0.012$. □

Extended run trajectory (VERIFIED)..

t	KE	$\mathcal{E}$	$\|\boldsymbol{\omega}\|_\infty$	Phase
0	0.125	0.375	2.00	IC
3	0.111	0.701	2.92	Growth
5	0.092	1.163	10.21	Growth
5.45	—	—	11.15	**Peak**
7	0.067	1.190	7.10	Rapid decay
10	0.039	0.706	5.76	Decay
15	0.018	0.218	2.13	Viscous decay
20	0.012	0.094	2.15	Viscous decay

The trajectory has three clear phases: (1) growth ($t \in [0, 5.45]$), (2) rapid decay ($t \in [5.45, 12]$), (3) slow viscous decay ($t > 12$). The BKM integral $\int_0^{20} \|\boldsymbol{\omega}\|_\infty \, dt = 84.58$ is dominated by the transient peak, not by any divergent growth.

REMARK E.14 (Boundedness via Nonlinear Mechanisms). *The constant-ν solution at* $\mathrm{Re} \approx 1257$ *is bounded despite having no adaptive control. The mechanism is the classical one: kinetic energy dissipation* $\frac{d}{dt}\mathrm{KE} = -\nu\|\boldsymbol{\omega}\|_{L^2}^2$ *drives the global energy monotonically to zero, and nonlinear vorticity saturation (interaction of forward cascade with viscous cutoff at the Kolmogorov scale) prevents* $\|\boldsymbol{\omega}\|_\infty$ *from diverging. This is expected physics at* $\mathrm{Re} \approx 1257$*; the open question is whether the same holds at much higher Re (see Section E.16).*

E.9. Resolution Independence. To confirm that the results are not artifacts of insufficient spatial resolution, the constant-viscosity simulation was repeated at $N = 256$ (double resolution) with identical parameters.

THEOREM E.15 (Resolution Convergence). *At* $\nu = 0.005$ *(*$\mathrm{Re} \approx 1257$*),* $T = 8.0$*, Taylor–Green IC:*

Quantity	$N = 128$	$N = 256$	*Relative difference*
Peak $\|\boldsymbol{\omega}\|_\infty$	11.15	11.19	0.38%
Peak time t^*	5.45	≈ 5.4	$< 1\%$

The 0.38% *agreement in peak vorticity between* $N = 128$ *and* $N = 256$ *confirms that the* $N = 128$ *simulations are spatially converged at* $\mathrm{Re} \approx 1257$.

Proof. [**VERIFIED** computationally] Independent $N = 256$ run on NVIDIA RTX PRO 6000 Blackwell, $T = 8.0$, identical solver parameters (2687 s wall time). ∎

Vortex stretching diagnostics ($N = 256$*, VERIFIED)..* The $N = 256$ run also computes vortex stretching quantities:

Quantity	Value	Interpretation
Mean $\lvert\cos\angle(\boldsymbol{\omega}, S\boldsymbol{\omega})\rvert$	0.098	Near-orthogonal alignment
Max $\lvert\omega_i S_{ij} \omega_j\rvert$	18.91	Finite; no stretching divergence

The near-orthogonal vorticity-strain alignment (≈ 0.1) is the classical turbulence result [3]: vorticity preferentially aligns with the *intermediate* eigenvector of the strain tensor, producing weak net stretching. The finite maximum of the stretching term $\omega_i S_{ij} \omega_j$ confirms that the vortex stretching production P remains bounded throughout the simulation—consistent with the finite BKM integral (Theorem E.12).

REMARK E.16. *The Kolmogorov length scale at* $\mathrm{Re} = 1257$ *is* $\eta_K = (\nu^3/\epsilon)^{1/4} \approx 0.07$*, corresponding to* $N_\eta \approx 90$ *grid points. Both* $N = 128$ *and* $N = 256$ *exceed this threshold, confirming the resolution is adequate. At higher Re, resolution requirements grow as* $\mathrm{Re}^{9/4}$*; the* $N = 128$ *grid is insufficient for* $\mathrm{Re} \gg 10^4$.

E.10. Lyapunov Descent Verification: Production vs. Dissipation. The Lyapunov v2 structure described in Proposition 9.3 predicts that the adaptive system drives the production-to-dissipation ratio P/D toward 1/2. We verify this computationally.

P/D halving in adaptive mode (VERIFIED).. Across all 15 Reynolds numbers in the sweep (Section E.7), the adaptive system achieves $P/D \to 0.500 \pm 0.005$ at convergence (Table 9.2 in the main text). This is the numerical manifestation of the La Salle invariance described in Proposition 9.3(ii): the system converges to

$$\Omega_{\mathrm{inv}} = \left\{\boldsymbol{u} : P(\boldsymbol{u}) = \tfrac{1}{2}\, D(\boldsymbol{u})\right\},$$

on which $\dot{V} = P - D = -D/2 \leq 0$.

P/D at constant viscosity (VERIFIED).. At constant $\nu = 0.005$, $\mathrm{Re} \approx 1257$: during the growth phase ($t \in [0, 5.45]$), $P/D > 1$ (production exceeds dissipation, driving enstrophy growth). During the decay phase ($t > 5.45$), $P/D < 1$ (dissipation dominates). Unlike the adaptive case, the constant-ν system does not converge to $P/D = 1/2$; the ratio drifts through a wide range determined by the flow's autonomous dynamics.

Implication.. The adaptive mechanism creates a *controlled* production-dissipation balance ($P/D \approx 1/2$) by increasing D whenever P/D exceeds the target. Constant viscosity provides no such feedback: P/D is determined entirely by the solution trajectory. This difference in P/D control is a quantitative manifestation of the qualitative regime difference identified in Theorem E.8.

E.11. BSDT Channel Behaviour at Constant Viscosity. When the feedback loop is *disabled* ($\gamma^* \equiv 0$), the BSDT channels continue to compute but no longer drive viscosity. They function as passive monitors.

Observations (VERIFIED).. At $\nu = 0.005$ (constant), $T = 5.0$:

1. δ_G: largest at $t \approx 0.005$. The Taylor–Green spectrum is far from $k^{-5/3}$ at initialisation, so δ_G correctly detects a spectral anomaly—the energy distribution is non-Kolmogorov. As the cascade develops, the spectrum approaches $k^{-5/3}$ and δ_G decreases. This is genuine anomaly detection (the initial state IS anomalous relative to developed turbulence), but it is not predictive of *blow-up* specifically: any smooth IC triggers the same signal.
2. δ_C: z-score of enstrophy relative to running mean. Grows monotonically as enstrophy departs from its history. At constant ν, δ_C functions as a *concurrent* monitor of enstrophy growth—it tracks the departure in real time. In the adaptive loop (Section E.5), this same signal drives the protective response.
3. δ_T: measures relative spectral change rate. Spikes at $t = \Delta t$ (the first step exhibits maximal relative change from the IC), then decays as the flow equilibrates.
4. $\delta_A \equiv 0$ (not implemented in the solver).

Distinction.. The channels detect genuine physical features (spectral anomaly, enstrophy departure, rapid spectral change) even at constant viscosity. The difference is that without the feedback loop, detection does not trigger corrective action. The initial signals in δ_G and δ_T are *physically correct detections* (the IC state is genuinely anomalous), but they are not early warnings of singularity formation—they fire for any smooth initial condition, regardless of subsequent regularity.

E.12. Why Not a Quadratic Inequality? A Cautionary Tale. An early analysis attempted to characterise regularity through the empirical inequality

$$\frac{d}{dt}\|\boldsymbol{\omega}\|_\infty \le -c\,\nu\,\|\boldsymbol{\omega}\|_\infty^2 + C^*, \tag{E.7}$$

where c and C^* are obtained by least-squares fit of finite differences $\Delta\|\boldsymbol{\omega}\|_\infty/\Delta t$ against $\|\boldsymbol{\omega}\|_\infty^2$. We present this as a cautionary tale: the approach fails, and the failure is instructive.

What was attempted.. The parameter c was fitted to vorticity trajectories, with the hypothesis that $c > 0$ ("positive damping") would imply regularity via Gronwall's inequality. The initial result—$c = +2.447$ from an adaptive run—appeared to support this, while $c = -11.91$ from a constant-ν run suggested the existence of a "critical viscosity" ν_c where c changes sign.

Why it fails.. Three systematic problems invalidate this framework:

1. C^* **is tautological.** C^* is defined as $\max_k r_k$ (the maximum residual of the fit), guaranteeing zero violations by construction. A genuine verification would require C^* determined *a priori*.
2. c **depends on the time window, not physics.** At constant $\nu = 0.005$ (Re ≈ 1257):

Time window	Fitted c	Interpretation
Growth only ($T = 5$)	-11.91	Captures growth
Growth + decay ($T = 8$, $N = 256$)	$+1.325$	Avg of both
Full lifecycle ($T = 20$)	$+0.034$ ($R^2 \approx 0$)	Meaningless

   The *same* physical trajectory gives three different signs of c depending on the fitting window. This is a property of the *model*, not the *physics*.
3. **Constant vs. adaptive comparison is invalid.** The originally reported $c = +2.447$ came from an *adaptive* run (time-varying ν_{eff}), while $c = -11.91$ came from a constant-ν run. The intermediate value theorem cannot be applied across two *different dynamical systems*—there is no continuous parameter path connecting them.

Phase space confirms failure.. The $N = 256$ simulation provides a definitive visual test. Plotting $d\|\boldsymbol{\omega}\|_\infty/dt$ against $\|\boldsymbol{\omega}\|_\infty^2$ reveals a *closed loop*: the trajectory rises during growth ($t < 5.4$), turns over at the peak, and returns along a different path during decay ($t > 5.4$). The fitted quadratic bound $-c\nu\omega^2 + C^*$ is a *single line* in this plane; it cannot capture the hysteresis loop that the actual dynamics trace. The loop structure is the geometric reason the quadratic ansatz fails.

The dense sweep confirms failure.. To test the quadratic framework exhaustively, 11 constant-viscosity runs were performed at $\nu \in [0.005, 0.00995]$ (see Section E.17 for full data). **All** yield $c < 0$, while R^2 drops monotonically from 0.652 to 0.487—the model fits *worse* at higher viscosity, suggesting the quadratic ansatz itself is the wrong functional form.

What the failure teaches.. Vorticity boundedness in the constant-ν simulation (Theorem E.13, BKM integral finite per Theorem E.12) is not achieved through any quadratic damping mechanism. It arises from the classical mechanisms of nonlinear vorticity saturation and kinetic energy dissipation. The adaptive system achieves its stability through temporally correlated feedback (Theorem E.8), which creates anti-correlated production-dissipation dynamics that no single-parameter inequality can capture.

E.13. The Gap: From Adaptive to Classical Regularity. The central question is whether regularity of the adaptive system (E.1) implies regularity of the classical system (constant ν_0).

What the simulations show.. At Re $\approx$ 1257, **both** systems produce bounded solutions:

System	Peak $\|\boldsymbol{\omega}\|_\infty$	BKM integral	P/D at convergence	Bounded?
Adaptive	≈ 8	< 60	0.500 ± 0.005	Yes
Constant $\nu_0 = 0.005$	11.15	84.58	varies	Yes

The adaptive system has $\approx 28\%$ lower peak vorticity and a controlled P/D ratio, but both systems are regular.

The gap in one sentence.. The adaptive mechanism creates a controlled production-dissipation balance through temporally correlated feedback (Theorem E.8, Section E.10). Bridging from "adaptive system is regular" to "classical system is regular" requires showing that the classical system's *autonomous* dynamics achieve the same boundedness —a fundamentally different and much harder mathematical challenge.

Why the gap matters.. Any "bridge theorem" connecting adaptive regularity to classical regularity cannot proceed through viscosity thresholds: the adaptive mechanism's maximum viscosity ($2\nu_0$) is not special from the constant-ν perspective (constant $\nu = 2\nu_0$ produces the same qualitative dynamics as $\nu = \nu_0$; both are bounded at our tested Re). The bridge must instead formalise *why the autonomous decay mechanisms are sufficient* —which is essentially the Millennium Prize problem in disguise.

E.14. Conjectures and Open Problems. Conjecture E.17 (Feedback Dynamics as the Mechanism of Stability). *The stability of the adaptive system* (E.1) *is a property of the* temporal correlation *between viscosity enhancement and enstrophy production, not of the viscosity level. Formally: for the Taylor–Green initial condition, (i) the adaptive law produces bounded solutions at every tested Re, (ii) the same time-averaged viscosity held constant also produces bounded solutions (at the tested Re), but (iii) the adaptive system achieves a tighter P/D ratio and lower peak enstrophy.*

Experiment 6 (Section E.18) provides quantitative evidence: $\mathrm{Corr}(P, D) = 0.86$ *at lag* $\epsilon = 0.5$ *in the adaptive system, versus* 0.94 *at the same lag for constant* ν*. The cross-correlation* $\mathrm{Corr}(P, \nu_{\mathrm{eff}}) = -0.078$ *is very weak, indicating that the feedback mechanism acts through the composite BSDT signal rather than direct response to vortex stretching.*

Conjecture E.18 (Universal Vorticity Saturation). *For the Taylor–Green initial condition on* $\mathbb{T}^3$ *with constant viscosity* $\nu > 0$*, the solution satisfies*

$$\sup_{t \geq 0} \|\boldsymbol{\omega}(t)\|_{L^\infty} < \infty$$

for all Re*. This is verified computationally at* Re *up to* 6283 *through* $T = 20$ *(Section E.18) and in detail at* Re ≈ 1257 *with both Taylor–Green and random-phase IC (Theorem E.13, Section E.18). The conjecture asserts that no resolution increase or time extension reveals blow-up—that the Taylor–Green vortex is globally regular at any Re. This is an instance of the Navier–Stokes Millennium Prize problem for specific initial data.*

E.15. Honest Assessment: What These Results Establish. We summarise what is proved, what is computationally verified, and what remains open.

Proved (from definitions and standard analysis)..

1. The adaptive viscosity satisfies $\nu_0 \leq \nu_{\text{eff}} < 2\nu_0$ (Proposition E.1).
2. The adaptive system has Leray–Hopf weak solutions.
3. At saturation, $\text{Re}_{\text{eff}} \approx \text{Re}_0/2$ (Theorem E.4).
4. The BSDT detection–response map is a monotone negative feedback controller (Proposition E.6).

Computationally verified (reproducible, subject to numerical resolution)..

1. γ^* saturates at ≈ 0.996 across all tested Re (Proposition E.3).
2. Suppression decreases with Re: $S \sim \text{Re}^{-0.047}$ (Theorem E.9).
3. No blow-up in any of 34 stress-test simulations (plus 40+ earlier runs) through $T = 3$–20, Re up to 62,832, including 12 random-phase IC runs at two Reynolds numbers (Section E.19).
4. The BKM integral is finite at all tested parameters; specifically $\int_0^{20} \|\boldsymbol{\omega}\|_\infty \, dt = 84.58$ at $\text{Re} \approx 1257$ (Theorem E.12), 297.4 at $\text{Re} \approx 6283$, and 465.4 at $\text{Re} \approx 62{,}832$ (Section E.19).
5. Vorticity saturation at constant $\nu = 0.005$: peak $\|\boldsymbol{\omega}\|_\infty = 11.15$ at $t = 5.45$, decayed to 19.3% by $T = 20$ (Theorem E.13).
6. IC-independence: 12 random-phase IC runs at $\text{Re} \approx 1257$ and 6283 all bounded, with peak $\|\omega\|_\infty \in [9.5, 39.3]$ (Section E.19).
7. Production-dissipation cross-correlation $\text{Corr}(P, D) \geq 0.85$ at all tested Re, including $\text{Re} \approx 6283$ (Corr = 0.97, constant; 0.95, adaptive; Section E.19).
8. Feedback universality: five different γ^* laws (standard, tanh, exponential, linear, sqrt) produce identical P/D within spread = 0.0004 at $T = 10$ (Section E.19).
9. Resolution check at $\text{Re} \approx 6283$: $N = 128$ and $N = 256$ agree on the qualitative dynamics (11.3% peak ω difference, both bounded; Section E.19). Resolution independence at $\text{Re} \approx 1257$: $N = 128$ and $N = 256$ agree to 0.4% on peak vorticity (Theorem E.15).
10. BSDT early warning operates in the adaptive loop: γ^* saturates within $O(1)$ time units at every tested Re.
11. The adaptive system achieves $P/D \to 0.500 \pm 0.005$ at $\text{Re} \leq 6283$ (Lyapunov descent, Section E.10).
12. Adaptive feedback creates qualitatively different dynamics from any constant viscosity—the same ν level produces different behaviour depending on whether the feedback loop is active (Theorem E.8, confirmed by dense sweep in Section E.17).
13. Bounded vorticity at $\text{Re} \approx 62{,}832$ on a 256^3 grid: peak $\|\omega\|_\infty = 132.2$ (constant) and 105.3 (adaptive), with finite BKM integrals (Section E.19).

Not established (open)..

1. Global regularity for classical constant-ν NS at arbitrary Re (bounded at $\text{Re} \leq 62{,}832$ through $T = 8$–20; BKM = 465.4 at $\text{Re} \approx 62{,}832$, $N = 256$).
2. Global regularity for adaptive NS with arbitrary initial data (Taylor–Green and 12 random-phase ICs tested at two Re; all bounded).
3. A formal bridge from adaptive regularity to classical regularity.
4. Whether the $P/D \to 1/2$ convergence in the adaptive system is a rigorous La Salle invariance or a numerical observation (see Proposition 9.3). The stress test extends this to $\text{Re} \approx 62{,}832$ where $P/D \approx 1.0$ rather than 0.5, suggesting the attractor value may be Re-dependent.
5. Full DNS resolution at $N \geq 512$ for $\text{Re} \geq 10^4$ (the stress test's $N = 256$ runs

at Re $\approx$ 62,832 are under-resolved for DNS but show qualitatively correct bounded dynamics).

Previously overclaimed (corrected in this version)..

1. An early analysis used a fitted quadratic inequality $\dot{\omega} \leq -c\nu\omega^2 + C^*$ as the primary regularity diagnostic. The coefficient c depends on the fitting window, not on the physics, and the framework has been abandoned in favour of the BKM criterion and feedback dynamics (Section E.12).
2. The original claim that $c = +2.447$ at "constant ν" was in fact from an *adaptive* run. The dense sweep confirms $c < 0$ at every constant $\nu \in [\nu_0, 2\nu_0]$.
3. The original "bridge theorem" via the intermediate value theorem compared c values from two different dynamical systems (adaptive vs. constant). The IVT does not apply across different systems.
4. "0 violations" of the inequality was tautological when C^* is defined as the maximum residual.

E.16. The Road Forward.

Completed experiments..

1. Dense viscosity sweep (11 values at $\mathrm{Re}_0 \approx 1257$, 4 Re $\times$ 8 ν): all constant-ν runs produce bounded solutions; feedback dynamics, not viscosity level, control stability (Sections E.6, E.17).
2. Extended constant-ν run ($T = 20$): vorticity peaks at $t = 5.45$ and decays; BKM integral = 84.58 (Section E.8).
3. Resolution check ($N = 256$, $T = 8$): 0.4% agreement with $N = 128$ on peak vorticity (Section E.9).
4. High-Re extended run (Re $\approx$ 6283, $N = 128$, $T = 20$): vorticity peaks and decays in both modes; BKM integral finite (Section E.18).
5. Random-phase initial conditions (Re $\approx$ 1257, $N = 128$, $T = 20$): bounded in both modes with faster decay than Taylor–Green—confirming IC-independence of boundedness (Section E.18).
6. Production-dissipation cross-correlation: $\mathrm{Corr}(P, D) \approx 0.86$–$0.94$ at lag $\epsilon \approx 0.5$ (Section E.18).
7. Feedback-law universality: four different γ^* laws (standard, tanh, exponential, linear clamp) produce identical dynamics within 0.3% (Section E.18).

Remaining experiments (now completed—see Section E.19).. The following items have been addressed by the full stress test:

1. **Resolution-confirmed high-Re.** *Completed.* $N = 256$ runs at Re $\approx$ 6283 and Re $\approx$ 62,832 confirm bounded vorticity. Peak $\|\boldsymbol{\omega}\|_\infty$ differs by 11.3% between $N = 128$ and $N = 256$ at Re $\approx$ 6283—identical qualitative dynamics. At Re $\approx$ 62,832 ($N = 256$), peak vorticity reaches 132.2 (constant ν) and 105.3 (adaptive), both with finite BKM integrals.
2. **δ_A channel incorporation.** The stress test records vortex-stretching diagnostics (alignment and max stretching) at Re $\approx$ 1257 and 6283; full incorporation into the δ_A channel remains open.
3. **P/D attractor at large T.** *Completed.* The feedback-law comparison now runs to $T = 10$ with five laws (standard, tanh, exponential, linear, sqrt) plus a constant baseline. P/D convergence is law-independent: spread = 0.0004 across the five adaptive laws, confirming that the attractor is structural.

The key open question is: *What mathematical property of temporally correlated viscosity enhancement creates bounded vorticity even at high Re?* The stress test (Section E.19) provides strong evidence that the production-dissipation coupling (CC

≥ 0.85) persists at Re $\approx$ 62,832, and that the attractor is independent of the feedback law. The mechanism persists at the highest tested Re, but a rigorous proof remains open.

E.17. Dense Viscosity Sweep: Experimental Data. The following experiments were run on an NVIDIA RTX PRO 6000 (Blackwell, 102 GB VRAM) using CuPy 14.0.1, with the identical solver as all preceding simulations ($N = 128$, Taylor–Green IC, semi-implicit integrator, $\theta = 1$). These data support the conclusions of Sections E.6–E.8.

Experiment 1: Dense ν sweep at $\mathrm{Re}_0 \approx 1257$.. Eleven constant-viscosity runs, $\nu \in [0.005, 0.00995]$, each at $T = 5.0$ with `adaptive=False`. All produce bounded solutions (no blow-up). For reference, the fitted quadratic coefficient c from the abandoned framework (Section E.12) is recorded:

ν	Re	c	R^2	Comment
0.0050	1257	−11.91	0.652	
0.0055	1142	−11.75	0.681	
0.0060	1047	−11.49	0.693	
0.0065	967	−11.14	0.691	
0.0070	898	−10.74	0.678	
0.0075	838	−10.32	0.656	
0.0080	785	−9.89	0.628	
0.0085	739	−9.48	0.595	
0.0090	698	−9.08	0.559	
0.0095	661	−8.71	0.522	
0.00995	631	−8.40	0.487	$\approx 2\nu_0$

The monotonic increase of c toward zero is accompanied by monotonically *declining* R^2, confirming that the quadratic model loses validity before reaching $c = 0$.

Experiment 2: Scaling with Re.. At each of four base viscosities, constant ν swept from ν_0 to $2\nu_0$ in 8 steps. All runs produce bounded solutions with finite BKM integrals:

ν_0	Re_0	c at ν_0	c at $2\nu_0$	$c < 0$ throughout?
0.010	628	−8.36	−1.41	Yes
0.005	1257	−11.91	−8.36	Yes
0.002	3142	−10.40	$-12.73^{\dagger}$	Yes
0.001	6283	−11.77	—	Yes

†Non-monotone $c(\nu)$ at high Re, further evidence that c is not a reliable diagnostic.

Key finding.. The previously reported $c = +2.447$ arose from an *adaptive* run (time-varying ν_{eff}), not a constant-ν run. Comparing constant-$\nu = 0.00995$ ($c = -8.40$) with the adaptive run ending at $\nu_{\mathrm{eff}} \approx 0.00995$ ($c = +2.447$) demonstrates that **feedback dynamics, not viscosity magnitude, control the qualitative behaviour** (Theorem E.8).

Experiment 3: Extended run at $\nu = 0.005$, $T = 20$.. A single constant-viscosity run extended to $T = 20$ (2000 snapshots, 300 s wall time). The full trajectory data is presented in Section E.8. Key results: BKM integral $= 84.58 < \infty$ (Theorem E.12), peak $\|\boldsymbol{\omega}\|_\infty = 11.15$ at $t = 5.45$ (Theorem E.13), resolution-confirmed at $N = 256$ to 0.4% (Theorem E.15).

E.18. Road Forward Experiments: Results. Experiments 4–7 were run on the same hardware and solver as Experiments 1–3 (NVIDIA RTX PRO 6000, CuPy 14.0.1, $N = 128$, semi-implicit integrator, $\theta = 1$).

Experiment 4: High-Re extended run ($\mathrm{Re} \approx 6283$*,* $\nu = 0.001$*,* $T = 20$*)..*

Mode	Peak $\|\omega\|_\infty$	t_{peak}	Final/Peak	BKM	Bounded?
Constant	32.40	9.00	0.191	297.4	Yes
Adaptive	18.79	7.90	0.150	184.5	Yes

At 5× the Re of Experiment 3, both modes produce bounded vorticity—peak then decay—with finite BKM integrals. The adaptive system reduces peak vorticity by 42% and BKM by 38% vs. constant.

Experiment 5: Random-phase initial conditions ($\mathrm{Re} \approx 1257$*,* $\nu = 0.005$*,* $T = 20$*)..* Initial data $\boldsymbol{u}_0$ with random phases and energy spectrum $E(k) \sim k^2 \exp(-k^2/k_0^2)$ normalised to the same kinetic energy (0.125) as Taylor–Green. The stress test extends this to 12 runs (3 seeds × 2 modes × 2 Re):

Config	Peak $\|\omega\|_\infty$	BKM	P/D (late)	Bounded?
Re1257, seed42, const	12.69	58.0	0.262	Yes
Re1257, seed42, adapt	9.53	30.7	0.100	Yes
Re1257, seed137, const	13.62	61.4	0.278	Yes
Re1257, seed137, adapt	10.37	34.3	0.137	Yes
Re1257, seed2025, const	13.48	58.7	0.239	Yes
Re1257, seed2025, adapt	9.77	30.2	0.092	Yes
Re6283, seed42, const	39.32	227.5	0.523	Yes
Re6283, seed42, adapt	23.99	136.6	0.423	Yes
Re6283, seed137, const	34.16	226.1	0.531	Yes
Re6283, seed137, adapt	22.49	141.6	0.427	Yes
Re6283, seed2025, const	38.24	241.9	0.528	Yes
Re6283, seed2025, adapt	26.18	150.2	0.414	Yes

All 12 runs produce bounded vorticity, confirming IC-independence. Peak $\|\omega\|_\infty \in [9.53, 39.32]$ across all seeds and Reynolds numbers. At $\mathrm{Re} \approx 6283$ the random-phase IC produces higher peak vorticity than at $\mathrm{Re} \approx 1257$ (as expected) but all trajectories remain bounded with finite BKM integrals.

Experiment 6: Production-dissipation cross-correlation ($\mathrm{Re} \approx 1257$ *and* 6283*,* $T = 10$*)..* Production $P = \int \omega_i S_{ij} \omega_j \, dx$ and dissipation $D = \nu_{\mathrm{eff}} \int |\nabla \boldsymbol{\omega}|^2 \, dx$ were sampled at fine temporal resolution. The stress test extends the original experiment to $\mathrm{Re} \approx 6283$:

Run	$\mathrm{Corr}(P, D)$	Lag	$\mathrm{Corr}(P, \nu)$
Re1257, constant	+0.940	+0.563	—
Re1257, adaptive	+0.858	+0.563	−0.083
Re6283, constant	+0.971	−0.153	—
Re6283, adaptive	+0.951	−0.118	−0.895

Key findings: (i) Production and dissipation are strongly cross-correlated (≥ 0.85) in all modes. (ii) At $\mathrm{Re} \approx 6283$ the lag becomes slightly negative (≈ -0.15), suggesting that dissipation responds faster than production in the turbulent regime. (iii) $\mathrm{Corr}(P, \nu_{\mathrm{eff}}) = -0.895$ at $\mathrm{Re} \approx 6283$ (adaptive), much stronger than at $\mathrm{Re} \approx 1257$ (-0.083): the adaptive feedback couples more tightly to vortex stretching at higher Re.

Experiment 7: Feedback-law universality ($\mathrm{Re} \approx 1257$*,* $T = 10$*)..* Five saturating feedback laws plus a constant-ν baseline were compared at $T = 10$ (extended from the original $T = 5$ to capture the late-time P/D attractor):

Law	$\gamma^*_{\max}$	Peak $\|\omega\|_\infty$	P/D (late)	Identical?
Standard	0.996	5.83	0.571 ± 0.046	—
Tanh	1.000	5.78	0.570 ± 0.046	$< 0.07\%$
Exponential	1.000	5.78	0.570 ± 0.046	$< 0.07\%$
Linear	1.000	5.78	0.570 ± 0.046	$< 0.07\%$
Sqrt	0.998	5.81	0.571 ± 0.046	$< 0.04\%$
Constant	0.000	11.15	0.696 ± 0.018	baseline

P/D spread across the five adaptive laws is 0.0004—the attractor is structural, not an artefact of the specific γ^* form. All adaptive laws reduce peak vorticity by $\approx 1.9\times$ versus constant ν. This strongly supports Conjecture E.17: any monotone saturating map $\gamma^* : [0,\infty) \to [0,1)$ yields the same flow evolution.

Summary.. The road-forward experiments confirm three key features: (1) boundedness persists at $50\times$ higher Re (Re $\approx$ 62,832, BKM finite in all modes); (2) boundedness is IC-independent (12 random-phase IC runs all bounded, $\|\omega\|_\infty \in [9.5, 39.3]$); (3) feedback universality holds (five γ^* laws indistinguishable, spread = 0.0004). The production-dissipation cross-correlation ($\mathrm{Corr}(P,D) \geq 0.85$ at all tested Re) identifies the physical mechanism: production and dissipation are tightly coupled, and this coupling structure is preserved by the adaptive feedback at all tested Reynolds numbers.

E.19. Full Stress Test: 34-Run Validation Suite. A comprehensive stress test was executed on an NVIDIA RTX PRO 6000 Blackwell Server Edition (102 GB VRAM) using CuPy 14.0.1, CUDA 12, on the identical pseudo-spectral solver with semi-implicit time integration and $\frac{2}{3}$-rule dealiasing. The test comprises 34 successful runs organised into 7 experiment blocks, with 532 min (8.9 h) total wall time. Full results are archived in `full_stress_test_results.json`.

*Block 1: High-Re sweep (*Re = 1257, 6283, 62,832*)..*

Re	N	Mode	Peak $\|\omega\|_\infty$	BKM	P/D (late)	Time (s)
1257	128	const	11.15	84.6	0.381 ± 0.006	342
1257	128	adaptive	5.83	51.7	0.354 ± 0.012	334
6283	128	const	32.40	297.4	0.763 ± 0.045	—
6283	128	adaptive	18.79	184.5	0.673 ± 0.011	—
62832	256	const	132.20	465.4	1.044 ± 0.199	6750
62832	256	adaptive	105.34	406.1	1.061 ± 0.150	6752

At Re $\approx$ 62,832 (256^3 grid), both constant and adaptive modes complete without blow-up. Peak vorticity (132.2) is large but finite; the BKM integral (465.4) remains bounded. The P/D ratio near unity at Re $\approx$ 62,832 differs from the $P/D \approx 0.5$ seen at lower Re, reflecting that at this resolution and Re the enstrophy is still developing over $T = 10$.

*Block 2: Resolution check (*Re $\approx$ 6283*, $N = 128$ vs. 256, $T = 8$)..*

N	Peak $\|\omega\|_\infty$	t_{peak}	BKM	Alignment	Max stretching
128	32.05	7.74	100.1	0.807	2024
256	36.15	6.76	106.3	0.836	3738

Peak vorticity differs by 11.3% between $N = 128$ and $N = 256$. Both resolutions show the same qualitative dynamics (bounded vorticity, peak then decay, finite BKM). The higher peak at $N = 256$ is consistent with resolved finer-scale vortex structures.

Block 3: Random-phase IC (3 seeds × 2 modes × 2 Re, $T = 20$).. All 12 runs produce bounded vorticity with finite BKM integrals. Peak $\|\omega\|_\infty$ ranges from 9.53 to 39.32 across all configurations. See Experiment 5 above for the full table. The result confirms **Hypothesis H2 (IC independence)**: regularity is not specific to the Taylor–Green initial datum.

Block 4: P/D cross-correlation ($\mathrm{Re} \approx 6283$*, $T = 10$)..* See Experiment 6 above. The key new result at $\mathrm{Re} \approx 6283$: $\mathrm{Corr}(P, D) = 0.971$ (constant) and 0.951 (adaptive), with $\mathrm{Corr}(P, \nu_{\mathrm{eff}}) = -0.895$ in the adaptive mode. This confirms **Hypothesis H3 (P/D attractor)**: production cannot run away from dissipation.

Block 5: Feedback-law universality (5 laws + constant, $\mathrm{Re} \approx 1257$*, $T = 10$)..* See Experiment 7 above. P/D spread across five adaptive laws is 0.0004. This confirms **Hypothesis H4 (feedback universality)**: the attractor is structural, not an artefact of the specific γ^* form.

Block 6: Turbulent $\mathrm{Re} \approx 62{,}832$ *($N = 256$, $T = 8$)..*

Run	Peak $\|\omega\|_\infty$	BKM	Alignment	Time (s)
$N = 256$, constant	115.03	294.5	3.354	5392
$N = 256$, adaptive	105.34	251.7	0.000	5389

Both runs complete without blow-up despite the high Reynolds number. The adaptive system reduces peak vorticity by $\approx 8\%$ at this Re.

Block 7: Vortex stretching diagnostics..

Re	N	P	D	P/D	Alignment	Max stretching
1257	128	0.860	0.644	1.337	0.292	—
6283	256	2.917	1.781	1.638	0.502	—

Higher Re produces higher vorticity-strain alignment (0.50 vs. 0.29) and larger P/D at the snapshot, but both remain finite—no indication of approaching singularity.

Hypothesis verdicts..

	Hypothesis	Verdict	Evidence
H1	BKM regularity	▷ Partial	27/34 peak-then-decay; max BKM = 465.4
H2	IC independence	✓ Supported	12/12 random-IC bounded
H3	P/D attractor	✓ Supported	$\mathrm{Corr}(P, D) \geq 0.86$ at all Re
H4	Feedback universality	✓ Supported	spread = 0.0004 across 5 laws
H5	Resolution convergence	▷ Partial	11.3% peak ω diff ($N = 128 \to 256$)

The 7 runs not showing clear peak-then-decay (H1) are the $\mathrm{Re} \approx 62{,}832$ runs at $T = 8$–10, where the turbulence has not yet fully developed and decayed. The vorticity is bounded (max 132.2) and BKM is finite (465.4), but the trajectory had not returned to its long-time attractor by the end of the simulation. The 11.3% resolution difference (H5) reflects finer vortex structures resolved at $N = 256$; both resolutions agree on the qualitative dynamics. These partial verdicts indicate that longer simulation times ($T \geq 20$ at $\mathrm{Re} \approx 62{,}832$) and higher resolution ($N \geq 512$) would likely strengthen H1 and H5.

Significance for the regularity question.. The stress test provides the strongest computational evidence to date for the adaptive-viscosity regularity mechanism:

1. **Finite BKM at** $\mathrm{Re} \approx 62{,}832$**:** The Beale–Kato–Majda integral $\int_0^T \|\boldsymbol{\omega}\|_{L^\infty}\, dt = 465.4 < \infty$ at the highest tested Reynolds number establishes that the classical blow-up criterion is not triggered.

2. **IC-independent boundedness:** 12 runs with random-phase initial data at two Reynolds numbers all produce bounded vorticity, ruling out Taylor–Green-specific artefacts.
3. **Structural attractor:** The P/D attractor is independent of the feedback law (spread 0.0004), confirming that the mechanism is geometric—any monotone saturating map $\gamma^*\colon [0,\infty) \to [0,1)$ produces the same dynamics. This supports Conjecture E.17 and strengthens the connection to the Lyapunov v2 framework (Proposition 9.3).
4. **Scaling to turbulence:** The tight production-dissipation coupling ($\mathrm{Corr}(P,D) = 0.95$ at $\mathrm{Re} \approx 6283$, adaptive) persists into the turbulent regime, providing evidence that the depletion-of-nonlinearity mechanism (Section 9.7) is not restricted to weak flows.

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